# Ridge Regression, Gradient Descent, And Perceptron

## CSAI 253 - Spring 2024

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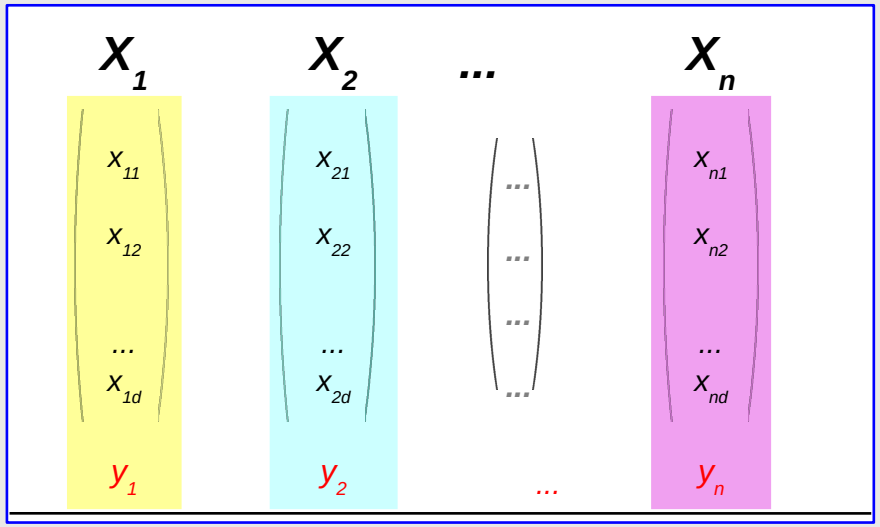
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# Revision of Standard Linear Regression Model

# Matrix Notation for Linear Regression and Classification (N- Samples)



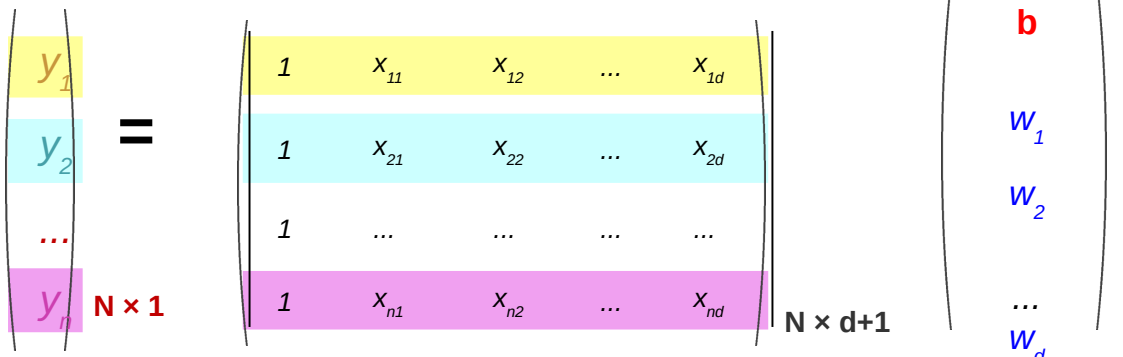
**N Training Samples**




***Remember***

$$\underline{y} = \underline{W}^T\underline{X} = \underline{X}^T\underline{W}$$

In General: **N >>> d**




$\underline{Y} = \underline{X}\,\underline{W}$

Where: $\underline{X}_{N\times d+1}$ **is a matrix**

$\underline{Y}_{N\times 1}$ **is labels vector**

# Concept of LOSS Function



$$Let\ \underline{\mathbf{Y}} = \begin{pmatrix} y_1 \\ y_2 \\ \dots \\ y_n \end{pmatrix}_{N \times 1}$$

**Original Labels**

from Training Samples

$$Let\ \underline{\hat{\mathbf{Y}}} = \begin{pmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \dots \\ \hat{y}_n \end{pmatrix}_{N \times 1}$$

**Predicted Labels**

from Model $\hat{Y}=X\underline{W}$

**Remember**

$$\|a\|_2 = a^T a$$

Objective is $\hat{Y}=Y$

If $\hat{Y} \neq Y \Rightarrow$ There is an **Error (Loss)**

$\underline{\hat{\mathbf{Y}}} = X\ \underline{W}$ ***Predicted***

*Where* : $X_{N \times d+1}$ is a **matrix** ⇒ **Given**

$\underline{Y}_{N \times 1}$ is labels **vector** ⇒ **Given**

$\underline{W}$ is Model Parameter ⇒ **Required**

Define **Loss** function as **squared distance** between $\hat{Y}$ and $Y$

$$Loss = \|Y - \hat{\mathbf{Y}}\|_2 = (Y - \hat{\mathbf{Y}})^T (Y - \hat{\mathbf{Y}})$$

# Minimization of Loss function to obtain BEST model parameters



$$Loss = \frac{1}{2}(Y - \hat{Y})^T(Y - \hat{Y}) = \frac{1}{2}(Y - XW)^T(Y - XW)$$

$$Loss = \frac{1}{2}[Y^TY + (XW)^T(XW) - Y^T(XW) - (XW)^TY]$$

$$Loss = \frac{1}{2}[Y^TY + (XW)^T(XW) - 2(XW)^TY]$$

**Remember**

X is a matrix, W is a vector,

XW is a vector

$$Y^T(XW) = (XW)^TY$$

$\frac{\partial Loss}{\partial W} = 0$ Get Best **W** that minimizes the loss

$$\frac{\partial Loss}{\partial W} = 0 = \frac{1}{2}[(XW)^T(X) + (X)^T(XW) - 2(X)^TY]$$

**Remember**

$$X^T(XW) = (XW)^TX$$

$$\frac{\partial Loss}{\partial W} = 0 = \frac{1}{2}[2X^TXW - 2X^TY] \rightarrow (X^TX)\underline{W} = X^TY \rightarrow$$

$$W = (X^TX)^{-1}X^TY$$

# Problems of Linear Regression

*(With Linearly Separable Data)*

# Problems of Linear Regression

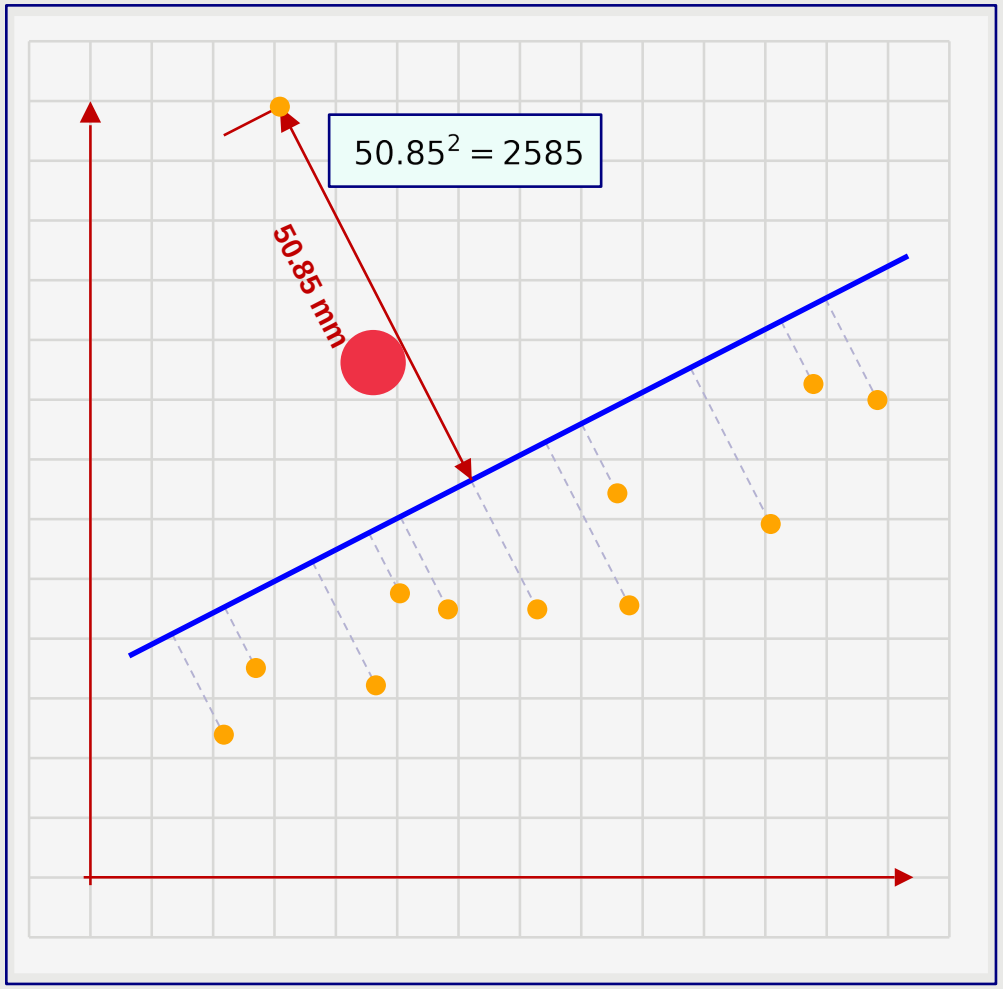
Assume the Dataset is linearly separable (suitable to apply Linear regression model)

**Linear regression model has FOUR problems:**

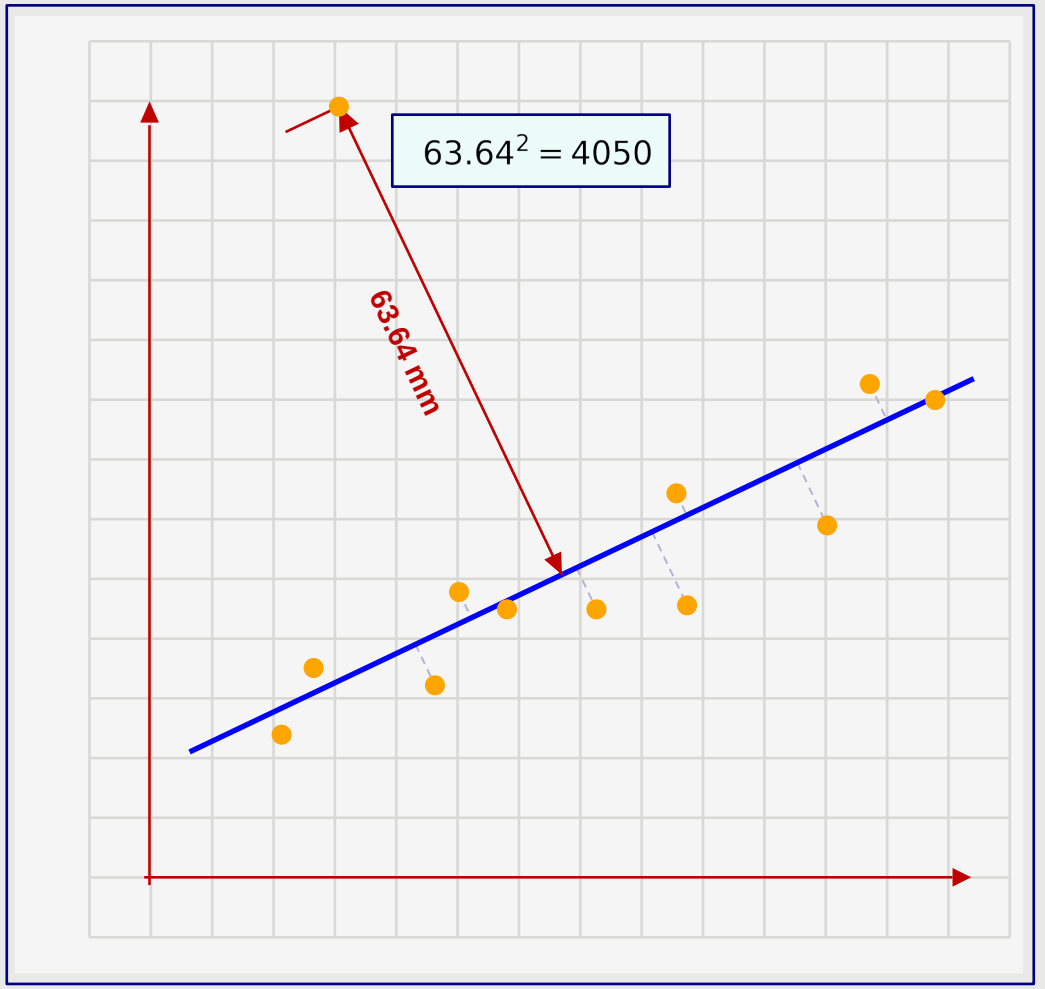
- **Overfitting Problem**
- **Matrix Inversion Problem [1]** ***(Singular Matrix)***
- **Matrix Inversion Problem [2]** ***(Large Matrix dimension)***
- **Different Hyperplanes can solve the problems** ***(Selection of Best Hyperplane)***

# [1] Ridge Regularizer to Minimize Overfitting

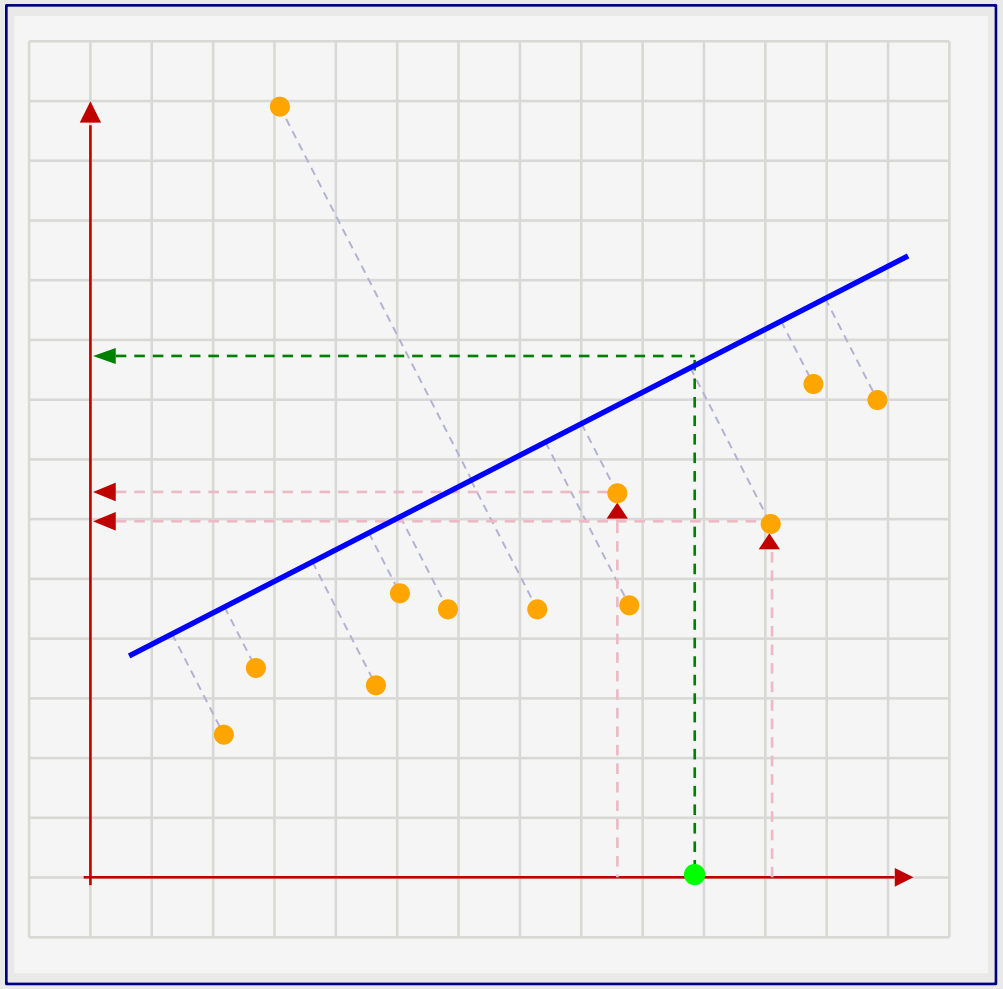
***(Also Known as L2 regularizer)***



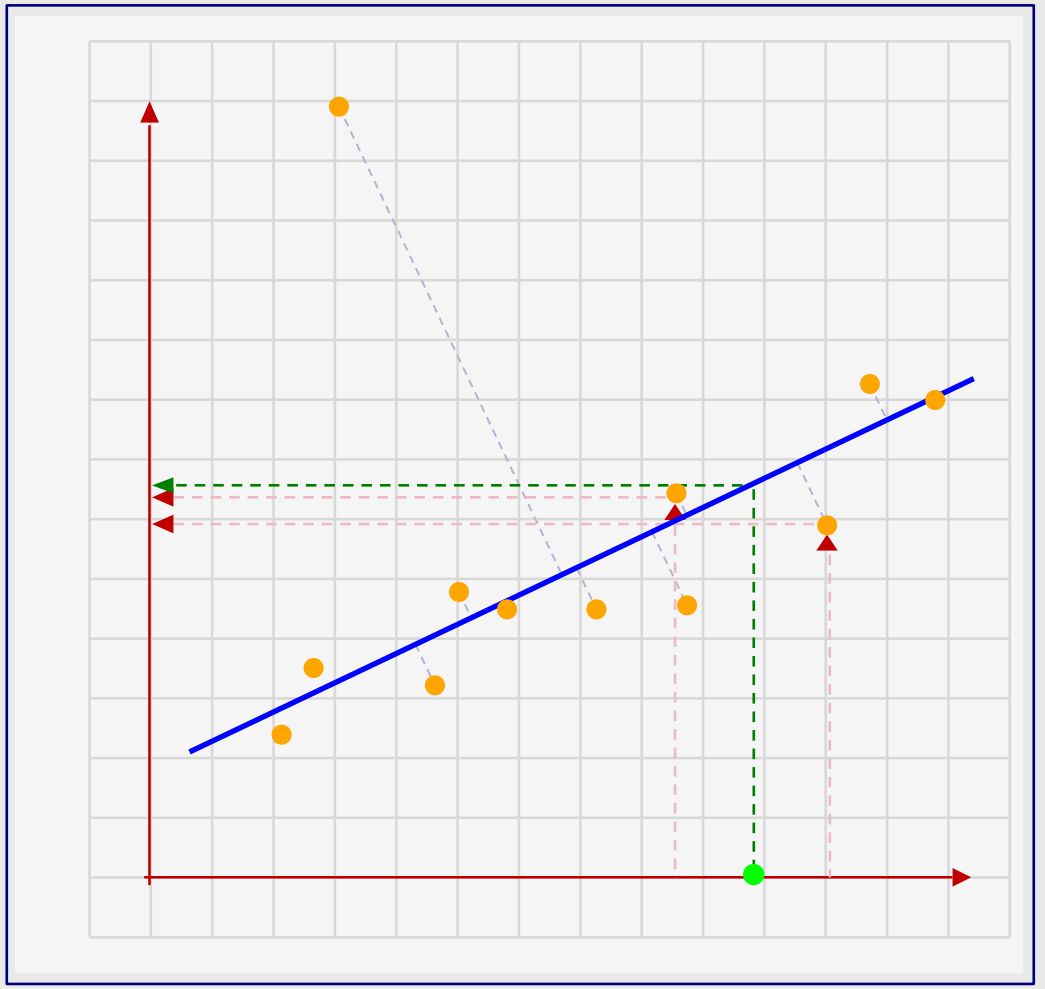

**Overfit** (try to minimize the residuals ONLY)
Fails to detect the Pattern




**Generalized** (Residuals are not minimized)
Successes to detect the Pattern



**Overfit** (try to minimize the residuals ONLY)
Fails to detect the Pattern



**Generalized** (Residuals are not minimized)
Successes to detect the Pattern

# Example of Overfitting due to outlier in data

*(and how to minimize overfitting )*

# Effect of Overfitting on model parameter “W”

**Training Data**

**Perfect Data**
**Y follows x1**

| | X1 | X2 | X3 | X4 | X5 | X6 | X7 | X8 |
|---|---|---|---|---|---|---|---|---|
| X1 | 1 | 2 | 3 | 4 | 8 | 9 | 10 | 11 |
| X2 | 1 | 3 | 5 | 6 | 16 | 7 | 5 | 3 |
| y | 2 | 4 | 6 | 8 | 16 | 18 | 20 | 22 |

**X5 is Outlier**

| | X1 | X2 | X3 | X4 | X5 | X6 | X7 | X8 |
|---|---|---|---|---|---|---|---|---|
| X1 | 1 | 2 | 3 | 4 | 8 | 9 | 10 | 30 |
| X2 | 1 | 3 | 5 | 6 | 16 | 7 | 5 | 3 |
| y | 2 | 4 | 6 | 8 | 16 | 18 | 20 | 22 |

**X5 is Outlier**

| | X1 | X2 | X3 | X4 | X5 | X6 | X7 | X8 |
|---|---|---|---|---|---|---|---|---|
| X1 | 1 | 2 | 3 | 4 | 8 | 9 | 10 | 60 |
| X2 | 1 | 3 | 5 | 6 | 16 | 7 | 5 | 3 |
| y | 2 | 4 | 6 | 8 | 16 | 18 | 20 | 22 |

**Test Data**

| | Test1 | Test2 | Test3 |
|---|---|---|---|
| X1 | 30 | 40 | 50 |
| X2 | 15 | 20 | 25 |
| y | 60 | 80 | 100 |

**Again**
**Y follows x1**

What will be the effect of X8 Outlier values on the trained model ???

# Effect of Overfitting on model parameter “W”



**Train**

| | $X_1$ | $X_2$ | $X_3$ | $X_4$ | $X_5$ | $X_6$ | $X_7$ | $X_8$ |
|---|---|---|---|---|---|---|---|---|
| $X_1$ | 1 | 2 | 3 | 4 | 8 | 9 | 10 | 11 |
| $X_2$ | 1 | 3 | 5 | 6 | 16 | 7 | 5 | 3 |
| y | 2 | 4 | 6 | 8 | 16 | 18 | 20 | 22 |

$$\underline{W} = \begin{pmatrix} 0 \\ 2 \\ 0 \end{pmatrix}$$

$$\|W\|^2 = 4$$

**Test**

| $Test_1$ | $Test_2$ | $Test_3$ |
|---|---|---|
| 30 | 40 | 50 |
| 15 | 20 | 25 |
| 60 | 80 | 100 |

| | $X_1$ | $X_2$ | $X_3$ | $X_4$ | $X_5$ | $X_6$ | $X_7$ | $X_8$ |
|---|---|---|---|---|---|---|---|---|
| $X_1$ | 1 | 2 | 3 | 4 | 8 | 9 | 10 | 30 |
| $X_2$ | 1 | 3 | 5 | 6 | 16 | 7 | 5 | 3 |
| y | 2 | 4 | 6 | 8 | 16 | 18 | 20 | 22 |

$$\underline{W} = \begin{pmatrix} 2.7 \\ 0.67 \\ 0.64 \end{pmatrix}$$

$$\|W\|^2 = 8.11$$

| $Test_1$ | $Test_2$ | $Test_3$ |
|---|---|---|
| 30 | 40 | 50 |
| 15 | 20 | 25 |
| 32.4 | 42.3 | 52.3 |

| | $X_1$ | $X_2$ | $X_3$ | $X_4$ | $X_5$ | $X_6$ | $X_7$ | $X_8$ |
|---|---|---|---|---|---|---|---|---|
| $X_1$ | 1 | 2 | 3 | 4 | 8 | 9 | 10 | 60 |
| $X_2$ | 1 | 3 | 5 | 6 | 16 | 7 | 5 | 3 |
| y | 2 | 4 | 6 | 8 | 16 | 18 | 20 | 22 |

$$\underline{W} = \begin{pmatrix} 4 \\ 0.29 \\ 0.78 \end{pmatrix}$$

$$\|W\|^2 = 16.7$$

| $Test_1$ | $Test_2$ | $Test_3$ |
|---|---|---|
| 30 | 40 | 50 |
| 15 | 20 | 25 |
| 24.4 | 31.2 | 38 |

# LOSS Function that MINIMIZES Overfitting

Let $\underline{Y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}_{N_{\times 1}}$

**Original Labels**

from Training Samples

Let $\hat{Y} = \begin{pmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \vdots \\ \hat{y}_n \end{pmatrix}_{N_{\times 1}}$

**Predicted Labels**

from Model $\hat{Y} = \underline{\mathbf{X}}\ \underline{\mathbf{W}}$

Objective is $\underline{\hat{Y}=Y}$ and $||W||$ is minimum This is called **Ridge Regularization**

Objective function **J(W)**=**Loss**(X,Y,W)+$\lambda$**R(W)** ($\lambda$ is a weighting factor)

Define **Loss** function as **squared distance** between $\hat{Y}$ and $Y$

**Loss Term** = $||Y-\hat{Y}||^2 = (Y-\hat{Y})^T(Y-\hat{Y})$

**Regularization Term** R = $||W||^2 = W^TW$

Objective function **J(W)** $J(W) = \frac{1}{2}[(Y-\hat{Y})^\top (Y-\hat{Y}) + \lambda W^\top W]$

## Minimization of Objective function to obtain BEST model parameters



$$V) = \frac{1}{2}(Y - \hat{Y})^T(Y - \hat{Y}) + \frac{1}{2}\lambda W^T W = \frac{1}{2}(Y - XW)^T(Y - XW) + \frac{1}{2}\lambda W^T W$$

$$J(W) = \frac{1}{2}[Y^T Y + (XW)^T(XW) - Y^T(XW) - (XW)^T Y] + \frac{1}{2}\lambda W^T W$$

$$J(W) = \frac{1}{2}[Y^T Y + (XW)^T(XW) - 2(XW)^T Y] + \frac{1}{2}\lambda W^T W$$

$\frac{\partial J(W)}{\partial W} = 0$ Get Best W that minimizes the Objective Function J(W)

**Remember**

$\frac{\partial W^T W}{\partial W} = 2W$

$$\frac{(W)}{W} = 0 = \frac{1}{2}[(XW)^T(X) + (X)^T(XW) - 2(X)^T Y] + \lambda W$$

$$\frac{\partial J(W)}{\partial W} = 0 = \frac{1}{2}[2X^T XW - 2X^T Y] + \lambda IW$$

$$(X^T X + \lambda I)W = X^T Y \rightarrow \underline{W} = (X^T X + \lambda I)^{-1} X^T Y$$

# $L_1$ and $L_2$ Regularization



**Regularizers are based on:**

**Penalize large values of W (model parameters)**

- This have the effect of better generalization of the model
- Since there is a contradiction between model training (using training dataset) and model generalization, the generalization term is added to the objective loss function with a customized weigh factor (do regularization to what extent)
- Penalization may be applied on the Squared Norm of the Weight Vector
  This is called $L_2$ regularization (also know as Ridge Regularization )
- Penalization may also be applied on the sum of Absolute values of the Weight Vector
  This is called $L_1$ regularization
- Regularization is particularly important for Small number of samples (N) and Large vector Length (d)

# Adding Ridge Regularizer to the model λ=3

**Train**

| | $X_1$ | $X_2$ | $X_3$ | $X_4$ | $X_5$ | $X_6$ | $X_7$ | $X_8$ |
|---|---|---|---|---|---|---|---|---|
| $X_1$ | 1 | 2 | 3 | 4 | 8 | 9 | 10 | 11 |
| $X_2$ | 1 | 3 | 5 | 6 | 16 | 7 | 5 | 3 |
| y | 2 | 4 | 6 | 8 | 16 | 18 | 20 | 22 |

$$\underline{W} = \begin{pmatrix} 0.1 \\ 1.95 \\ 0.03 \end{pmatrix}$$

$$\|W\|^2 = 3.8$$

**Test**

| | $Test_1$ | $Test_2$ | $Test_3$ |
|---|---|---|---|
| $X_1$ | 30 | 40 | 50 |
| $X_2$ | 15 | 20 | 25 |
| y | 59 | 78.7 | 98.4 |

| | $X_1$ | $X_2$ | $X_3$ | $X_4$ | $X_5$ | $X_6$ | $X_7$ | $X_8$ |
|---|---|---|---|---|---|---|---|---|
| $X_1$ | 1 | 2 | 3 | 4 | 8 | 9 | 10 | 30 |
| $X_2$ | 1 | 3 | 5 | 6 | 16 | 7 | 5 | 3 |
| y | 2 | 4 | 6 | 8 | 16 | 18 | 20 | 22 |

$$\underline{W} = \begin{pmatrix} 1.2 \\ 0.72 \\ 0.77 \end{pmatrix}$$

$$\|W\|^2 = 2.45$$

| | $Test_1$ | $Test_2$ | $Test_3$ |
|---|---|---|---|
| $X_1$ | 30 | 40 | 50 |
| $X_2$ | 15 | 20 | 25 |
| y | 34.1 | 45.1 | 56.1 |

| | $X_1$ | $X_2$ | $X_3$ | $X_4$ | $X_5$ | $X_6$ | $X_7$ | $X_8$ |
|---|---|---|---|---|---|---|---|---|
| $X_1$ | 1 | 2 | 3 | 4 | 8 | 9 | 10 | 60 |
| $X_2$ | 1 | 3 | 5 | 6 | 16 | 7 | 5 | 3 |
| y | 2 | 4 | 6 | 8 | 16 | 18 | 20 | 22 |

$$\underline{W} = \begin{pmatrix} 1.77 \\ 0.32 \\ 0.99 \end{pmatrix}$$

$$\|W\|^2 = 4.2$$

| | $Test_1$ | $Test_2$ | $Test_3$ |
|---|---|---|---|
| $X_1$ | 30 | 40 | 50 |
| $X_2$ | 15 | 20 | 25 |
| y | 26.2 | 34.4 | 42.5 |

# Adding Ridge Regularizer to the model λ=10



**Train**

| | $X_1$ | $X_2$ | $X_3$ | $X_4$ | $X_5$ | $X_6$ | $X_7$ | $X_8$ |
|---|---|---|---|---|---|---|---|---|
| $X_1$ | 1 | 2 | 3 | 4 | 8 | 9 | 10 | 11 |
| $X_2$ | 1 | 3 | 5 | 6 | 16 | 7 | 5 | 3 |
| y | 2 | 4 | 6 | 8 | 16 | 18 | 20 | 22 |

$$\underline{W} = \begin{pmatrix} 0.13 \\ 1.87 \\ 0.09 \end{pmatrix}$$

$$\|W\|^2 = 3.5$$

**Test**

| | $Test_1$ | $Test_2$ | $Test_3$ |
|---|---|---|---|
| $X_1$ | 30 | 40 | 50 |
| $X_2$ | 15 | 20 | 25 |
| y | 57.4 | 76.5 | 95.6 |

| | $X_1$ | $X_2$ | $X_3$ | $X_4$ | $X_5$ | $X_6$ | $X_7$ | $X_8$ |
|---|---|---|---|---|---|---|---|---|
| $X_1$ | 1 | 2 | 3 | 4 | 8 | 9 | 10 | 30 |
| $X_2$ | 1 | 3 | 5 | 6 | 16 | 7 | 5 | 3 |
| y | 2 | 4 | 30 | 8 | 16 | 18 | 20 | 22 |

$$\underline{W} = \begin{pmatrix} 0.54 \\ 0.73 \\ 0.81 \end{pmatrix}$$

$$\|W\|^2 = 1.48$$

| | $Test_1$ | $Test_2$ | $Test_3$ |
|---|---|---|---|
| $X_1$ | 30 | 40 | 50 |
| $X_2$ | 15 | 20 | 25 |
| y | 34.6 | 46.4 | 57.4 |

| | $X_1$ | $X_2$ | $X_3$ | $X_4$ | $X_5$ | $X_6$ | $X_7$ | $X_8$ |
|---|---|---|---|---|---|---|---|---|
| $X_1$ | 1 | 2 | 3 | 4 | 8 | 9 | 10 | 60 |
| $X_2$ | 1 | 3 | 5 | 6 | 16 | 7 | 5 | 3 |
| y | 2 | 4 | 60 | 8 | 16 | 18 | 20 | 22 |

$$\underline{W} = \begin{pmatrix} 0.82 \\ 0.33 \\ 1.1 \end{pmatrix}$$

$$\|W\|^2 = 1.92$$

| | $Test_1$ | $Test_2$ | $Test_3$ |
|---|---|---|---|
| $X_1$ | 30 | 40 | 50 |
| $X_2$ | 15 | 20 | 25 |
| y | 26.8 | 35.4 | 44.1 |

## Test Data

$z_1 = \begin{pmatrix} 30 \\ 15 \end{pmatrix}$ y=60 $z_2 = \begin{pmatrix} 40 \\ 20 \end{pmatrix}$ y=80 $z_3 = \begin{pmatrix} 50 \\ 25 \end{pmatrix}$ y=100

| Outlier $X_s$ | NO Regularizer with (as if λ=0) | Ridge Regularizer with λ=3 | Ridge Regularizer with λ=10 |
|---|---|---|---|
| $\begin{pmatrix} 11 \\ 3 \end{pmatrix}$ 22 | W = $\begin{pmatrix} 0 \\ 2 \\ 0 \end{pmatrix}$ ‖W‖²=4 | W = $\begin{pmatrix} 0.1 \\ 1.95 \\ 0.03 \end{pmatrix}$ ‖W‖²=3.8 | W = $\begin{pmatrix} 0.13 \\ 1.87 \\ 0.09 \end{pmatrix}$ ‖W‖²=3.5 |
| $\begin{pmatrix} 30 \\ 3 \end{pmatrix}$ 22 | W = $\begin{pmatrix} 2.7 \\ 0.67 \\ 0.64 \end{pmatrix}$ ‖W‖²=8.11 | W = $\begin{pmatrix} 1.2 \\ 0.72 \\ 0.77 \end{pmatrix}$ ‖W‖²=2.45 | W = $\begin{pmatrix} 0.54 \\ 0.73 \\ 0.81 \end{pmatrix}$ ‖W‖²=1.48 |
| $\begin{pmatrix} 60 \\ 3 \end{pmatrix}$ 22 | W = $\begin{pmatrix} 4 \\ 0.29 \\ 0.78 \end{pmatrix}$ ‖W‖²=16.7 | W = $\begin{pmatrix} 1.77 \\ 0.32 \\ 0.99 \end{pmatrix}$ ‖W‖²=4.2 | W = $\begin{pmatrix} 0.82 \\ 0.33 \\ 1.1 \end{pmatrix}$ ‖W‖²=1.92 |

| | Test₁ | Test₂ | Test₃ |
|---|---|---|---|
| $x_1$ | 30 | 40 | 50 |
| $x_2$ | 15 | 20 | 25 |
| y | 60 | 80 | 100 |

| | Test₁ | Test₂ | Test₃ |
|---|---|---|---|
| $x_1$ | 30 | 40 | 50 |
| $x_2$ | 15 | 20 | 25 |
| y | 59 | 78.7 | 98.4 |

| | Test₁ | Test₂ | Test₃ |
|---|---|---|---|
| $x_1$ | 30 | 40 | 50 |
| $x_2$ | 15 | 20 | 25 |
| y | 57.4 | 76.5 | 95.6 |

| | Test₁ | Test₂ | Test₃ |
|---|---|---|---|
| $x_1$ | 30 | 40 | 50 |
| $x_2$ | 15 | 20 | 25 |
| y | 32.4 | 42.3 | 52.3 |

| | Test₁ | Test₂ | Test₃ |
|---|---|---|---|
| $x_1$ | 30 | 40 | 50 |
| $x_2$ | 15 | 20 | 25 |
| y | 34.1 | 45.1 | 56.1 |

| | Test₁ | Test₂ | Test₃ |
|---|---|---|---|
| $x_1$ | 30 | 40 | 50 |
| $x_2$ | 15 | 20 | 25 |
| y | 34.6 | 46 | 57.4 |

| | Test₁ | Test₂ | Test₃ |
|---|---|---|---|
| $x_1$ | 30 | 40 | 50 |
| $x_2$ | 15 | 20 | 25 |
| y | 24.4 | 31.2 | 38 |

| | Test₁ | Test₂ | Test₃ |
|---|---|---|---|
| $x_1$ | 30 | 40 | 50 |
| $x_2$ | 15 | 20 | 25 |
| y | 26.2 | 34.4 | 42.5 |

| | Test₁ | Test₂ | Test₃ |
|---|---|---|---|
| $x_1$ | 30 | 40 | 50 |
| $x_2$ | 15 | 20 | 25 |
| y | 26.8 | 35.4 | 44.1 |

# [2] Ridge Regularization to overcome

# Inverse Problem of Singular Matrix

# Inverse Problem of Singular Matrix

$X^T X$

| | | |
|---|---|---|
| 14 | 10 | 15 |
| 10 | 40 | 46 |
| 20 | 80 | 92 |

$|X^T X| = 0$ ⇒ Singular Matrix ⇒ *No Inverse*

$(X^T X + \lambda I)$ $\lambda = 3$

| | | |
|---|---|---|
| 17 | 10 | 15 |
| 10 | 43 | 46 |
| 20 | 80 | 95 |

$|X^T X| = 5685$ ⇒ Non-Singular Matrix ⇒ *Inverse Exists*

$(X^T X + \lambda I)^{-1}$ ⇒

| | | |
|---|---|---|
| 0.07 | 0.04 | -0.03 |
| -0.01 | 0.23 | -0.11 |
| -0.01 | -0.2 | 0.11 |

# [3] Inverse of High Dimension (non-singular) Matrix

# Inverse of High Dimension Matrix

*Recall*

$$\underline{W} = (X^T X + \lambda I)^{-1} X^T \underline{Y}$$

if Feature vectors are of dimension d $\Rightarrow$ ($\underline{X}^T X$) is of size d+1 x d+1

The standard matrix inversion algorithm has a time complexity of $O(n^3)$, which means that the computation time increases rapidly with the size of the matrix (increasing cubically with the number of components).

For Example: if Features vector is of size 1000 then Inverse calculation is or order 1000,000,000 which is impractical

**Solution:**

*Instead of using **Least Squares** with Matrix inversion,*
*use **Gradient Descent** iterative solution*

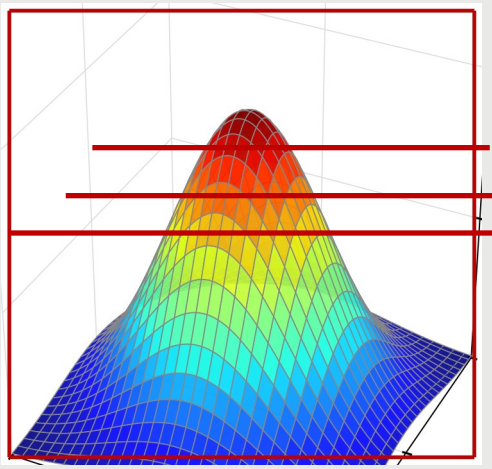
# Gradient Descent (GD)

## and

## How to apply to minimize Loss function

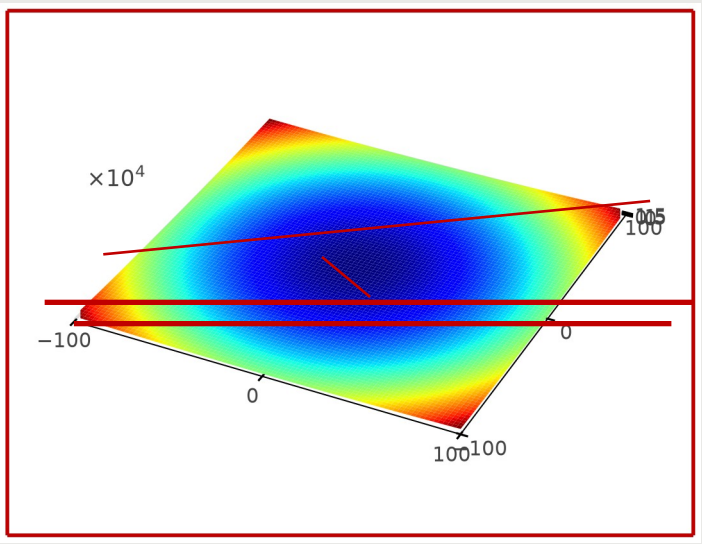
# Types of Loss function shapes

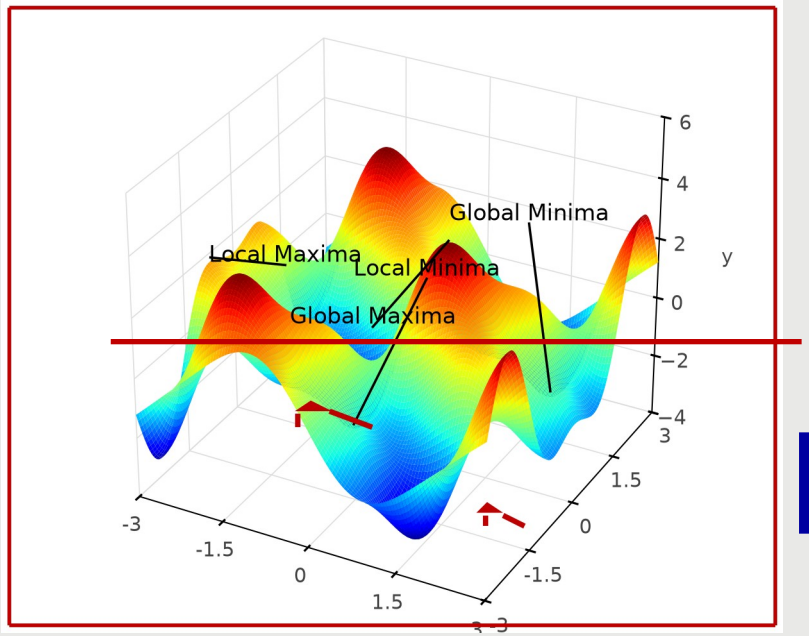




**Concave** Function
One Maxima

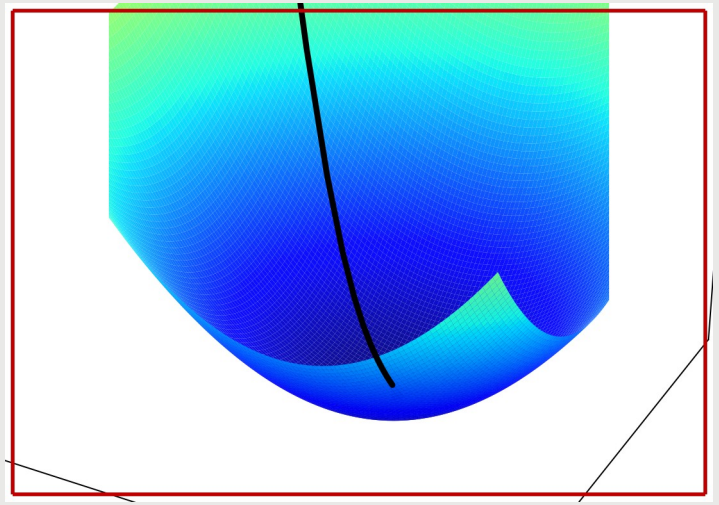
**Convex** Function
One Minima




**Non-Convex**
**Multiple Minima**

**Gradient Descent**
**Path to reach Minima**



# Gradient Descent Solution



**Gradient Descent (GD) Algorithm:**

[1] Start with initial **Random Weights (W)**

[2] Update the weights by moving along the direction of **negative gradient** of Loss function (w.r.t. W) (**moving step is a hyper parameter, AKA Learning Rate**)

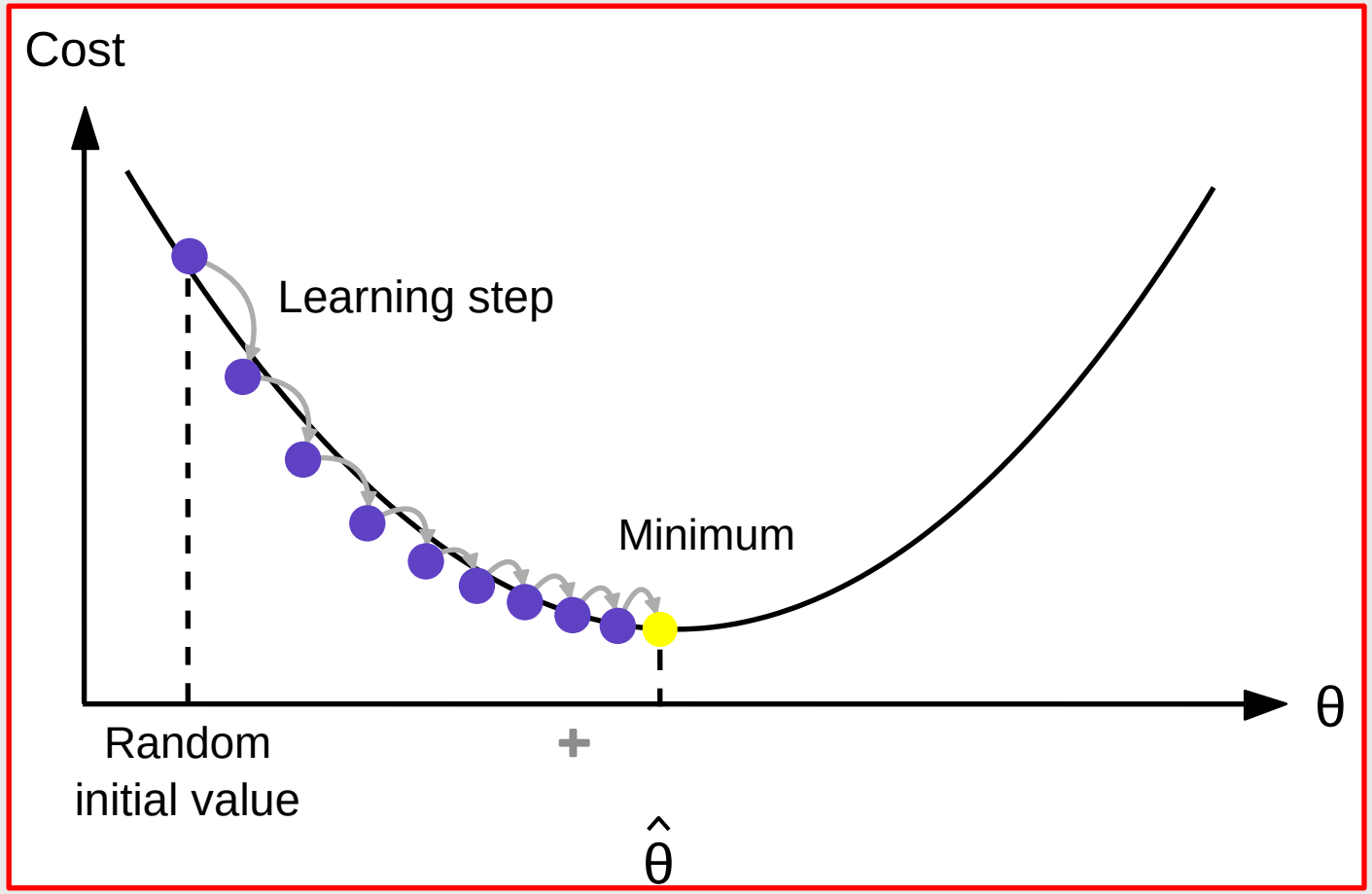
[3] Stop when the gradient is **almost zero** (this is at the minimum value of the function)

**Notes:**

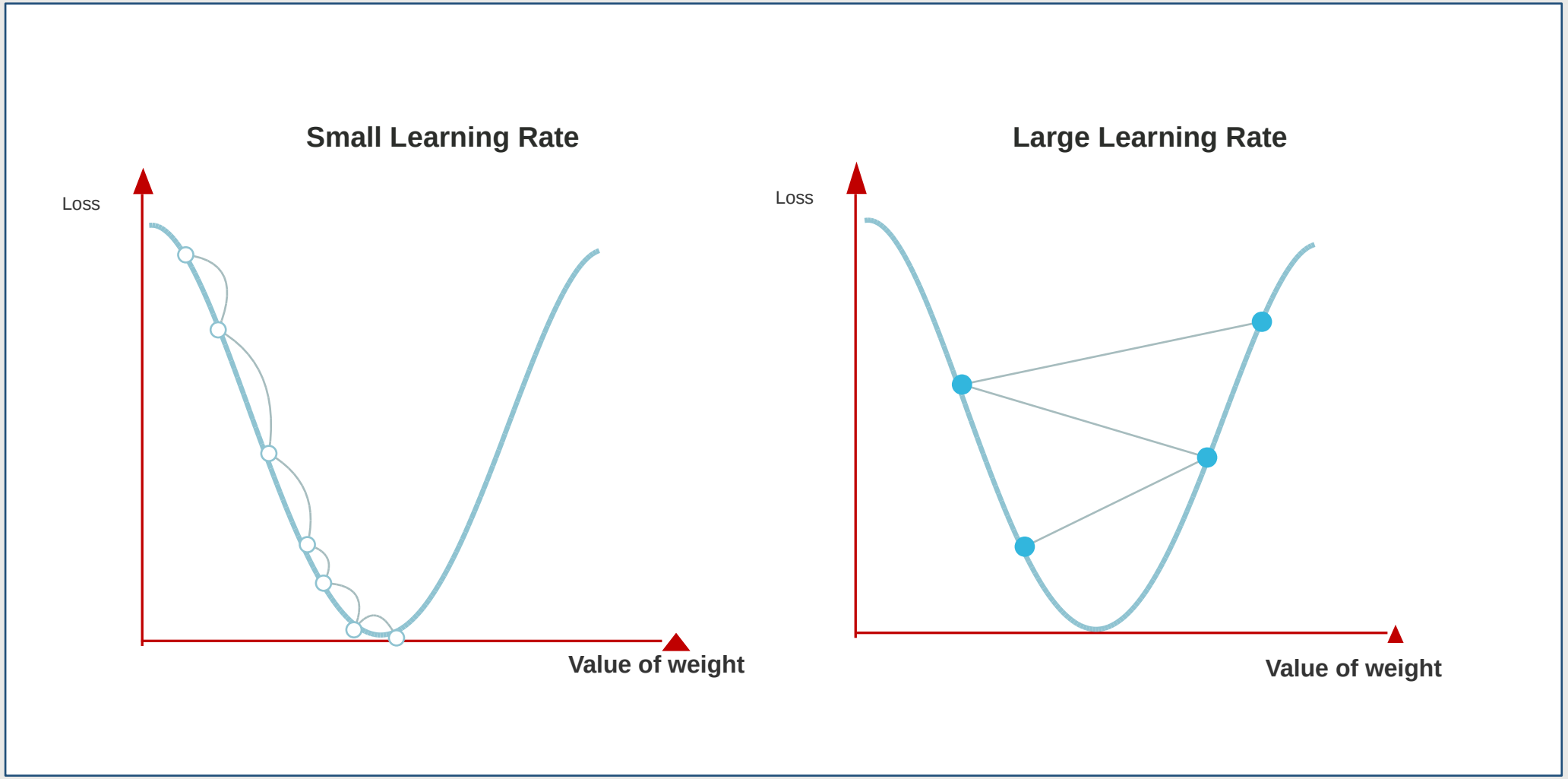
- **Least Squares** Linear regression objective function **is a convex function** (even with regularizer). This means it has **ONE Minimum** (good news ).

- GD will approximatly find this minimum value but at a variable number of steps depending on the used **Learning Rate**

# Follow Negative Direction of Gradient with Selected Learning Rate

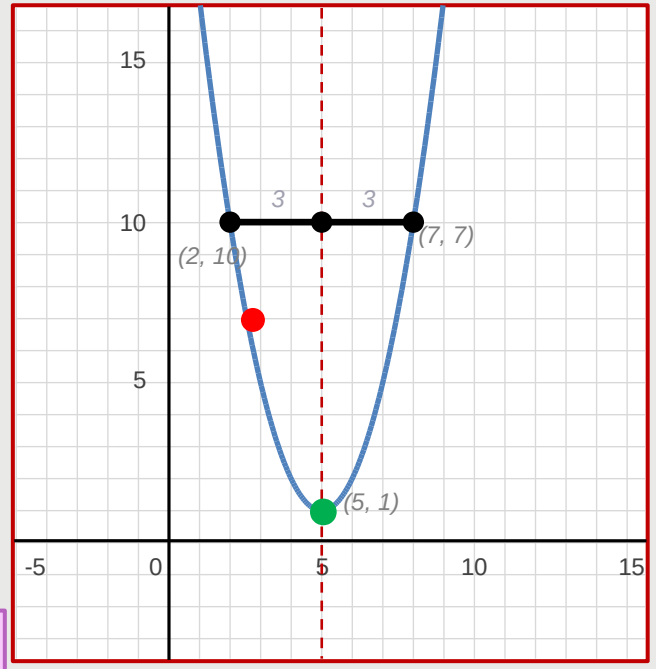


# Effect of small and large Learning Rate



# Example: Obtain Minimum Value of a function using Gradient Descent (GD)

# Convex Function, Learning Rate =0.2, Start at X=3




| step | 0.2 |
|---|---|
| initial X | 3 |
| delta | 0.1 |

$f(x) = (x-5)^2 + 1$

$X = 5$

$f(x)$ Minimum value is **1.0**

***Reach Minimum after 11 Steps***

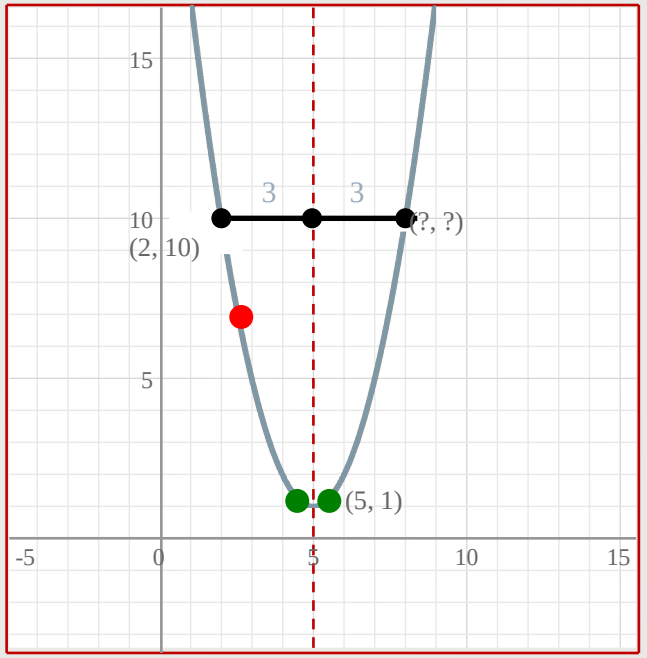
| x | 3 | 3.2 | 3.4 | 3.6 | 3.8 | 4 | 4.2 | 4.4 | 4.6 | 4.8 | 5 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| f(x) | 5 | 4.24 | 3.56 | 2.96 | 2.44 | 2 | 1.64 | 1.36 | 1.16 | 1.04 | 1 |
| x-delta | 2.9 | 3.1 | 3.3 | 3.5 | 3.7 | 3.9 | 4.1 | 4.3 | 4.5 | 4.7 | 4.9 |
| f(x-delta) | 5.41 | 4.61 | 3.89 | 3.25 | 2.69 | 2.21 | 1.81 | 1.49 | 1.25 | 1.09 | 1.01 |
| x+delta | 3.1 | 3.3 | 3.5 | 3.7 | 3.9 | 4.1 | 4.3 | 4.5 | 4.7 | 4.9 | 5.1 |
| f(x+delta) | 4.61 | 3.89 | 3.25 | 2.69 | 2.21 | 1.81 | 1.49 | 1.25 | 1.09 | 1.01 | 1.01 |
| slope | -8 | -7.2 | -6.4 | -5.6 | -4.8 | -4 | -3.2 | -2.4 | -1.6 | -0.8 | 0 |
| next x | 3.2 | 3.4 | 3.6 | 3.8 | 4 | 4.2 | 4.4 | 4.6 | 4.8 | 5 | stop |

# Convex Function, Learning Rate =0.3, Start at X=3



| step | 0.3 |
|---|---|
| initial X | 3 |
| delta | 0.1 |

$$f(x) = (x-5)^2 + 1$$




***X*** **is in between 4.8 and color red bold 5.1**
***f(x)*** **is in between 1.04 and 1.01**

**Bouncing**

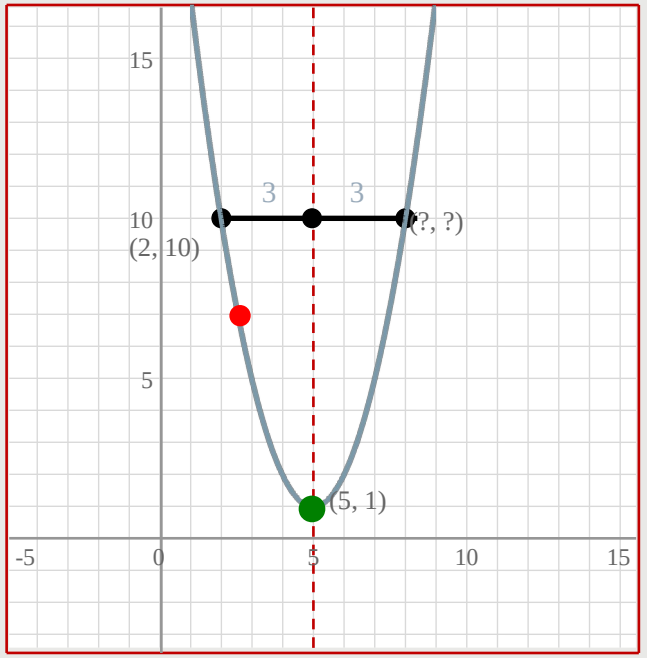
| | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| x | 3 | 3.3 | 3.6 | 3.9 | 4.2 | 4.5 | 4.8 | 5.1 | 4.8 | 5.1 | 4.8 | 5.1 | 4.8 | 5.1 | 4.8 | 5.1 | 4.8 | 5.1 |
| f(x) | 5 | 3.89 | 2.96 | 2.21 | 1.64 | 1.25 | 1.04 | 1.01 | 1.04 | 1.01 | 1.04 | 1.01 | 1.04 | 1.01 | 1.04 | 1.01 | 1.04 | 1.01 |
| x-delta | 2.9 | 3.2 | 3.5 | 3.8 | 4.1 | 4.4 | 4.7 | 5 | 4.7 | 5 | 4.7 | 5 | 4.7 | 5 | 4.7 | 5 | 4.7 | 5 |
| f(x-delta) | 5.41 | 4.24 | 3.25 | 2.44 | 1.81 | 1.36 | 1.09 | 1 | 1.09 | 1 | 1.09 | 1 | 1.09 | 1 | 1.09 | 1 | 1.09 | 1 |
| x+delta | 3.1 | 3.4 | 3.7 | 4 | 4.3 | 4.6 | 4.9 | 5.2 | 4.9 | 5.2 | 4.9 | 5.2 | 4.9 | 5.2 | 4.9 | 5.2 | 4.9 | 5.2 |
| f(x+delta) | 4.61 | 3.56 | 2.69 | 2 | 1.49 | 1.16 | 1.01 | 1.04 | 1.01 | 1.04 | 1.01 | 1.04 | 1.01 | 1.04 | 1.01 | 1.04 | 1.01 | 1.04 |
| slope | -8 | -6.8 | -5.6 | -4.4 | -3.2 | -2 | -0.8 | 0.4 | -0.8 | 0.4 | -0.8 | 0.4 | -0.8 | 0.4 | -0.8 | 0.4 | -0.8 | 0.4 |
| next x | 3.3 | 3.6 | 3.9 | 4.2 | 4.5 | 4.8 | 5.1 | 4.8 | 5.1 | 4.8 | 5.1 | 4.8 | 5.1 | 4.8 | 5.1 | 4.8 | 5.1 | 4.8 |

# Convex Function, Learning Rate =0.4, Start at X=3

| step | 0.4 |
|---|---|
| initial X | 3 |
| delta | 0.1 |

$$f(x) = (x-5)^2 + 1$$

***X* = 5**
*f(x)* Minimum value is **1.0**




***Reach Minimum after 6 Steps***

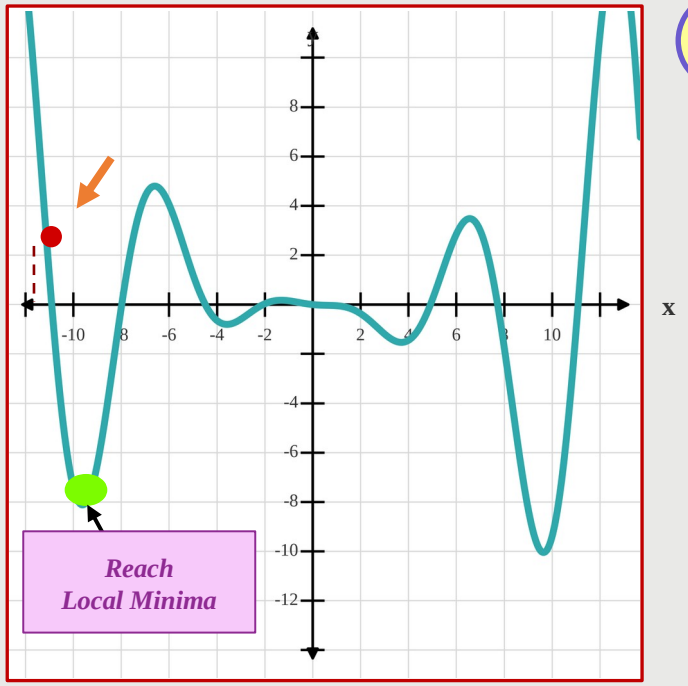
| x | 3 | 3.4 | 3.8 | 4.2 | 4.6 | 5 |
|---|---|---|---|---|---|---|
| f(x) | 5 | 3.56 | 2.44 | 1.64 | 1.16 | 1 |
| x-delta | 2.9 | 3.3 | 3.7 | 4.1 | 4.5 | 4.9 |
| f(x-delta) | 5.41 | 3.89 | 2.69 | 1.81 | 1.25 | 1.01 |
| x+delta | 3.1 | 3.5 | 3.9 | 4.3 | 4.7 | 5.1 |
| f(x+delta) | 4.61 | 3.25 | 2.21 | 1.49 | 1.09 | 1.01 |
| slope | -8 | -6.4 | -4.8 | -3.2 | -1.6 | 0 |
| next x | 3.4 | 3.8 | 4.2 | 4.6 | 5 | stop |

# Non-Convex Function

## May reach different minimum value based on:

## initial point and learning rate

$$f(x) = \frac{x^2\cos(x) - x}{10}$$




| step | 0.1 |
|---|---|
| initial X | -11 |
| delta | 0.1 |

***X*** is in between -9.7 and color red bold -9.6

***f(x)*** is in between -8.08 and -8.011

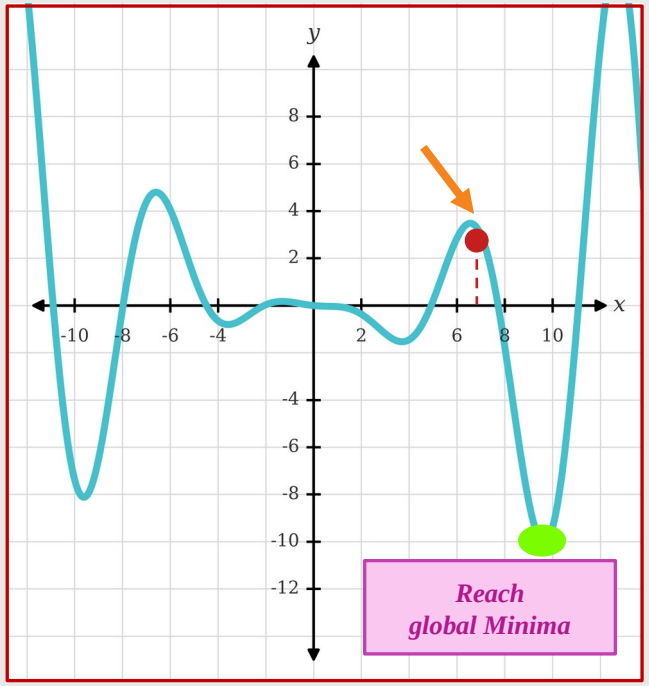
| x | -11 | -10.9 | -10.8 | -10.7 | -10.6 | -10.5 | -10.4 | -10.3 | -10.2 | -10.1 | -10 | -9.9 | -9.8 | -9.7 | -9.6 | -9.7 | -9.6 | -9.7 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| f(x) | 1.15 | -0.04 | -1.19 | -2.26 | -3.27 | -4.19 | -5.03 | -5.77 | -6.41 | -6.95 | -7.39 | -7.72 | -7.96 | -8.08 | -8.11 | -8.08 | -8.11 | -8.08 |
| x-delta | -11.1 | -11 | -10.9 | -10.8 | -10.7 | -10.6 | -10.5 | -10.4 | -10.3 | -10.2 | -10.1 | -10 | -9.9 | -9.8 | -9.7 | -9.8 | -9.7 | -9.8 |
| f(x-delta) | 2.39 | 1.15 | -0.04 | -1.19 | -2.26 | -3.27 | -4.19 | -5.03 | -5.77 | -6.41 | -6.95 | -7.39 | -7.72 | -7.96 | -8.08 | -7.96 | -8.08 | -7.96 |
| x+delta | -10.9 | -10.8 | -10.7 | -10.6 | -10.5 | -10.4 | -10.3 | -10.2 | -10.1 | -10 | -9.9 | -9.8 | -9.7 | -9.6 | -9.5 | -9.6 | -9.5 | -9.6 |
| f(x+delta) | -0.04 | -1.19 | -2.26 | -3.27 | -4.19 | -5.03 | -5.77 | -6.41 | -6.95 | -7.39 | -7.72 | -7.96 | -8.08 | -8.11 | -8.05 | -8.11 | -8.05 | -8.11 |
| slope | -24.3 | -23.4 | -22.2 | -20.8 | -19.3 | -17.6 | -15.8 | -13.8 | -11.8 | -9.8 | -7.7 | -5.7 | -3.6 | -1.5 | 0.3 | -1.5 | 0.3 | -1.5 |
| next x | -10.9 | -10.8 | -10.7 | -10.6 | -10.5 | -10.4 | -10.3 | -10.2 | -10.1 | -10 | -9.9 | -9.8 | -9.7 | -9.6 | -9.7 | -9.6 | -9.7 | -9.6 |

# Non-Convex Function

## May reach different minimum value based on:

## initial point and learning rate

$$f(x) = \frac{x^2\cos(x) - x}{10}$$




| step | 0.2 |
|---|---|
| initial X | 7 |
| delta | 0.1 |

***X*** is in between 9.6 and color red bold 9.8

***f(x)*** is in between -10.03 and -9.92

| x | 7 | 7.2 | 7.4 | 7.6 | 7.8 | 8 | 8.2 | 8.4 | 8.6 | 8.8 | 9 | 9.2 | 9.4 | 9.6 | 9.8 | 9.6 | 9.8 | 9.6 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| f(x) | 2.99 | 2.43 | 1.66 | 0.69 | -0.45 | -1.73 | -3.1 | -4.5 | -5.88 | -7.16 | -8.28 | -9.17 | -9.77 | -10.03 | -9.92 | -10.03 | -9.92 | -10.03 |
| x-delta | 6.9 | 7.1 | 7.3 | 7.5 | 7.7 | 7.9 | 8.1 | 8.3 | 8.5 | 8.7 | 8.9 | 9.1 | 9.3 | 9.5 | 9.7 | 9.5 | 9.7 | 9.5 |
| f(x-delta) | 3.19 | 2.74 | 2.07 | 1.2 | 0.14 | -1.08 | -2.41 | -3.8 | -5.2 | -6.54 | -7.75 | -8.76 | -9.51 | -9.95 | -10.02 | -9.95 | -10.02 | -9.95 |
| x+delta | 7.1 | 7.3 | 7.5 | 7.7 | 7.9 | 8.1 | 8.3 | 8.5 | 8.7 | 8.9 | 9.1 | 9.3 | 9.5 | 9.7 | 9.9 | 9.7 | 9.9 | 9.7 |
| f(x+delta) | 2.74 | 2.07 | 1.2 | 0.14 | -1.08 | -2.41 | -3.8 | -5.2 | -6.54 | -7.75 | -8.76 | -9.51 | -9.95 | -10.02 | -9.7 | -10.02 | -9.7 | -10.02 |
| slope | -4.5 | -6.7 | -8.7 | -10.6 | -12.2 | -13.3 | -13.9 | -14 | -13.4 | -12.1 | -10.1 | -7.5 | -4.4 | -0.7 | 3.2 | -0.7 | 3.2 | -0.7 |
| next x | 7.2 | 7.4 | 7.6 | 7.8 | 8 | 8.2 | 8.4 | 8.6 | 8.8 | 9 | 9.2 | 9.4 | 9.6 | 9.8 | 9.6 | 9.8 | 9.6 | 9.8 |

# Using Gradient Descent to train Linear Regression Model

## (Problem Formulation)

# Problem Formulation



Define two classes: class I and Class II

let $d(X)$ be a discriminant function = $W^T X$

if $X \in$ class I then $d(X) > 0$

if $X \in$ Class II then $d(X) < 0$

For misclassified $X$ :

If $X \in$ class I and misclassified as Class II then $d(X)$ will be $<0$

If $X \in$ Class II and misclassified as Class I then $d(X)$ will be $>0$

Define Error function as:

$$\text{Error} = \sum_{\text{all misclassified vectors}(i)} W^T X_i \quad \Rightarrow \quad (W^T X_i) \text{ take positive and negative values}$$ X

Define $\delta_i$ as :

If $X \in$ class I and misclassified as Class II then $d(X)$ will be $<0 \Rightarrow \delta_i = -1 \Rightarrow d(X)\, \delta_i > 0$

If $X \in$ Class II and misclassified as Class I then $d(X)$ will be $>0 \Rightarrow \delta_i = +1 \Rightarrow d(X)\, \delta_i > 0$

# Problem Formulation



**Define $\delta_i$ as :**

*If* $\underline{X} \in$ ***class I*** and ***misclassified*** as ***Class II*** *then* $d(\underline{X})$ *will be* $<0 \Rightarrow \delta_i = -1 \Rightarrow d(\underline{X})\delta_i > 0$

*If* $\underline{X} \in$ ***class II*** and ***misclassified*** as ***Class I*** *then* $d(\underline{X})$ *will be* $>0 \Rightarrow \delta_i = +1 \Rightarrow d(\underline{X})\delta_i > 0$

**Redefine Error function as:**

$$\text{Error} = \sum_{\text{all misclassified vectors (i)}} \delta_i \underline{W}^T \underline{X}_i \Rightarrow (\underline{W}^T \underline{X}_i) \text{ takes positive and negative values and } \delta_i \text{ has the same sign}$$

Summation is always ***positive***

*Minimize the Error using* ***Gradient Descent*** *with step* $\eta$

$$\underline{W}^{(updated)} = \underline{W}^{old} - \eta \frac{\partial Error}{\partial W} \quad \textbf{where} \quad \frac{\partial Error}{\partial W} = \sum_{\textbf{misclassified vectors}} \delta_i \underline{X}_i$$

$$\underline{W}^{(updated)} = \underline{W}^{old} - \eta \sum_{\textbf{misclassified vectors ONLY (i)}} \delta_i \underline{X}_i$$

# Gradient Descent (GD)

# vs

# Stochastic Gradient Descent (SGD)

# Gradient Descent (GD) – Stochastic Gradient Descent SGD) and Mini-Batch Gradient Descent

- Stochastic Gradient Descent (SGD) is a variant of GD which computes the gradient of the Error function with respect to a single training example
- In other words, it updates the weights using one training sample at a time
- Remember: GD updated the weight vector after passing through ALL training samples.

$$W^{(updated)} = W^{old} - \eta \delta_i X_i$$ Repeat for ALL misclassified Samples

**Why using SGD**

- SGD is usually more efficient than GD
- SGD converges faster than GD

**Batch GD is a generalization of GD and SGD where weights are updated after passing through "Mini-Batch" of training samples with size n**

**If n=N (it is GD), and if n=1 (it is SGD)**

# Training of Linear Regression Model

## usnig SGD

## Solved Example

$x_1$ $x_2$ $x_3$ $x_4$ $x_5$ $x_6$

$\begin{pmatrix}1\\1\end{pmatrix}$ $\begin{pmatrix}2\\1\end{pmatrix}$ $\begin{pmatrix}1\\2\end{pmatrix}$ $\begin{pmatrix}5\\4\end{pmatrix}$ $\begin{pmatrix}6\\5\end{pmatrix}$ $\begin{pmatrix}6\\6\end{pmatrix}$

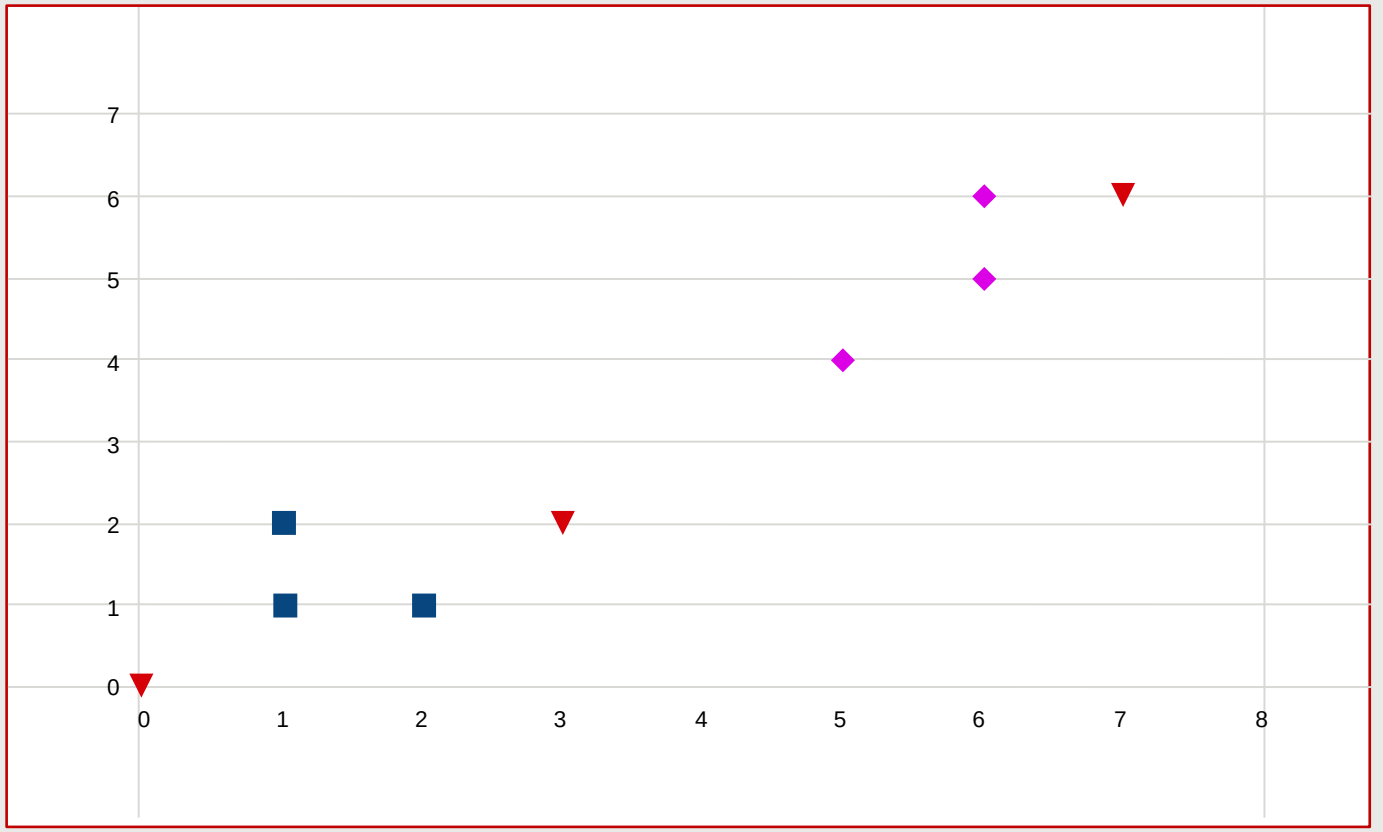
$C_1$ $C_1$ $C_1$ $C_2$ $C_2$ $C_2$

Initial W = $\begin{pmatrix}4\\-2\\-2\end{pmatrix}$ *Step η = 2*

$C_1$ positive, $C_2$ negative, Zero is unaccepted

Check classes of $z_1$ $z_1$ $z_1$

$\begin{pmatrix}0\\0\end{pmatrix}$ $\begin{pmatrix}7\\6\end{pmatrix}$ $\begin{pmatrix}3\\2\end{pmatrix}$



$x_1$ $x_2$ $x_3$ $x_4$ $x_5$ $x_6$

$\begin{pmatrix}1\\1\end{pmatrix}\begin{pmatrix}2\\1\end{pmatrix}\begin{pmatrix}1\\2\end{pmatrix}\begin{pmatrix}5\\4\end{pmatrix}\begin{pmatrix}6\\5\end{pmatrix}\begin{pmatrix}6\\6\end{pmatrix}$

$C_1$ $C_1$ $C_1$ $C_2$ $C_2$ $C_2$

Initial $\underline{W} = \begin{pmatrix}4\\-2\\-2\end{pmatrix}$ *Step η = 2*

$C_1$ positive $C_2$ negative Zero is unaccepted

Check classes of $\begin{pmatrix}0\\0\end{pmatrix}\begin{pmatrix}7\\6\end{pmatrix}\begin{pmatrix}3\\2\end{pmatrix}$ $Z_1$ $Z_1$ $Z_1$

**Ugmented Vectors ( from Top )**

$x_1$ $x_2$ $x_3$ $x_4$ $x_5$ $x_6$

$\begin{pmatrix}1\\1\\1\end{pmatrix}\begin{pmatrix}1\\2\\1\end{pmatrix}\begin{pmatrix}1\\1\\2\end{pmatrix}\begin{pmatrix}1\\5\\4\end{pmatrix}\begin{pmatrix}1\\6\\5\end{pmatrix}\begin{pmatrix}1\\6\\6\end{pmatrix}$

$C_1$ $C_1$ $C_1$ $C_2$ $C_2$ $C_2$

## Iteration 1

$d(x_1) = \begin{pmatrix}1&1&1\end{pmatrix}\begin{pmatrix}4\\-2\\-2\end{pmatrix} = 0$ ⇒ Update Weight $\underline{W} = \begin{pmatrix}4\\-2\\-2\end{pmatrix} + 2 * \begin{pmatrix}1\\1\\1\end{pmatrix} = \begin{pmatrix}6\\0\\0\end{pmatrix}$ ⇐

$d(x_2) = \begin{pmatrix}1&2&1\end{pmatrix}\begin{pmatrix}6\\0\\0\end{pmatrix} = 6$ ⇒ Keep same Weight $\underline{W} = \begin{pmatrix}6\\0\\0\end{pmatrix}$

$d(x_3) = \begin{pmatrix}1&1&2\end{pmatrix}\begin{pmatrix}6\\0\\0\end{pmatrix} = 6$ ⇒ Keep same Weight $\underline{W} = \begin{pmatrix}6\\0\\0\end{pmatrix}$

$d(x_4) = \begin{pmatrix}1&5&4\end{pmatrix}\begin{pmatrix}6\\0\\0\end{pmatrix} = 6$ ⇒ Update Weight $\underline{W} = \begin{pmatrix}6\\0\\0\end{pmatrix} - 2 * \begin{pmatrix}1\\5\\4\end{pmatrix} = \begin{pmatrix}4\\-10\\-8\end{pmatrix}$ ⇐

$d(x_5) = \begin{pmatrix}1&6&5\end{pmatrix}\begin{pmatrix}4\\-10\\-8\end{pmatrix} = -96$ ⇒ Keep same Weight $\underline{W} = \begin{pmatrix}4\\-10\\-8\end{pmatrix}$

$d(x_6) = \begin{pmatrix}1&6&6\end{pmatrix}\begin{pmatrix}4\\-10\\-8\end{pmatrix} = -104$ ⇒ Keep same Weight $\underline{W} = \begin{pmatrix}4\\-10\\-8\end{pmatrix}$

| $x_1$ | $x_2$ | $x_3$ | $x_4$ | $x_5$ | $x_6$ |
|---|---|---|---|---|---|
| (1) | (2) | (1) | (5) | (6) | (6) |
| (1) | (1) | (2) | (4) | (5) | (6) |
| $C_1$ | $C_1$ | $C_1$ | $C_2$ | $C_2$ | $C_2$ |

Initial $\underline{W} = \begin{pmatrix} 4 \\ -2 \\ -2 \end{pmatrix}$ ***Step η = 2***

Check classes of $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$ $\begin{pmatrix} 7 \\ 6 \end{pmatrix}$ $\begin{pmatrix} 3 \\ 2 \end{pmatrix}$ ($Z_1$ $Z_1$ $Z_1$)

$C_1$ positive $C_2$ negative Zero is unaccepted

**Ugmented Vectors ( from Top )**

| $x_1$ | $x_2$ | $x_3$ | $x_4$ | $x_5$ | $x_6$ |
|---|---|---|---|---|---|
| 1 | 1 | 1 | 1 | 1 | 1 |
| 1 | 2 | 1 | 5 | 6 | 6 |
| 1 | 1 | 2 | 4 | 5 | 6 |
| $C_1$ | $C_1$ | $C_1$ | $C_2$ | $C_2$ | $C_2$ |

## Iteration 2

$d(x_1) = \begin{vmatrix} 1 & 1 & 1 \end{vmatrix} \begin{pmatrix} 4 \\ -10 \\ -8 \end{pmatrix} = -14$ ⇒ Update Weight $\underline{W} = \begin{pmatrix} 4 \\ -10 \\ -8 \end{pmatrix} + 2 * \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 6 \\ -8 \\ -6 \end{pmatrix}$ ⇐

$d(x_2) = \begin{vmatrix} 1 & 2 & 1 \end{vmatrix} \begin{pmatrix} 6 \\ -8 \\ -6 \end{pmatrix} = -14$ ⇒ Update Weight $\underline{W} = \begin{pmatrix} 6 \\ -8 \\ -6 \end{pmatrix} + 2 * \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 8 \\ -4 \\ -4 \end{pmatrix}$ ⇐

$d(x_3) = \begin{vmatrix} 1 & 1 & 2 \end{vmatrix} \begin{pmatrix} 8 \\ -4 \\ -4 \end{pmatrix} = -4$ ⇒ Update Weight $\underline{W} = \begin{pmatrix} 8 \\ -4 \\ -4 \end{pmatrix} + 2 * \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 10 \\ -2 \\ 0 \end{pmatrix}$ ⇐

$d(x_4) = \begin{vmatrix} 1 & 5 & 4 \end{vmatrix} \begin{pmatrix} 10 \\ -2 \\ 0 \end{pmatrix} = 0$ ⇒ Update Weight $\underline{W} = \begin{pmatrix} 10 \\ -2 \\ 0 \end{pmatrix} - 2 * \begin{pmatrix} 1 \\ 5 \\ 4 \end{pmatrix} = \begin{pmatrix} 8 \\ -12 \\ -8 \end{pmatrix}$ ⇐

$d(x_5) = \begin{vmatrix} 1 & 6 & 5 \end{vmatrix} \begin{pmatrix} 8 \\ -12 \\ -8 \end{pmatrix} = -104$ ⇒ Keep same Weight $\underline{W} = \begin{pmatrix} 8 \\ -12 \\ -8 \end{pmatrix}$

$d(x_6) = \begin{vmatrix} 1 & 6 & 6 \end{vmatrix} \begin{pmatrix} 8 \\ -12 \\ -8 \end{pmatrix} = -112$ ⇒ Keep same Weight $\underline{W} = \begin{pmatrix} 8 \\ -12 \\ -8 \end{pmatrix}$

$x_1 \; x_2 \; x_3 \; x_4 \; x_5 \; x_6$

$\begin{pmatrix}1\\1\end{pmatrix} \begin{pmatrix}2\\1\end{pmatrix} \begin{pmatrix}1\\2\end{pmatrix} \begin{pmatrix}5\\4\end{pmatrix} \begin{pmatrix}6\\5\end{pmatrix} \begin{pmatrix}6\\6\end{pmatrix}$

$C_1 \; C_1 \; C_1 \; C_2 \; C_2 \; C_2$

Initial $W = \begin{vmatrix}4\\-2\\-2\end{vmatrix}$ **Step η = 2**

**C1** *positive,* **C2** *negative,* **Zero** is unaccepted

Check classes of $\underset{Z_1}{\begin{pmatrix}0\\0\end{pmatrix}} \underset{Z_1}{\begin{pmatrix}7\\6\end{pmatrix}} \underset{Z_1}{\begin{pmatrix}3\\2\end{pmatrix}}$

**Ugmented Vectors ( from Top )**

| $x_1$ | $x_2$ | $x_3$ | $x_4$ | $x_5$ | $x_6$ |
|---|---|---|---|---|---|
| (1) | (1) | (1) | (1) | (1) | (1) |
| (1) | (2) | (1) | (5) | (6) | (6) |
| (1) | (2) | (4) | (5) | (6) | (6) |
| $C_1$ | $C_1$ | $C_1$ | $C_2$ | $C_2$ | $C_2$ |

**Iteration 3**

$d(x_1) = \begin{pmatrix}1 & 1 & 1\end{pmatrix}\begin{pmatrix}8\\-12\\-8\end{pmatrix} = -12$ ⇒ Update Weight $\underline{W} = \begin{pmatrix}8\\-12\\-8\end{pmatrix} + 2 * \begin{pmatrix}1\\1\\1\end{pmatrix} = \begin{pmatrix}10\\-10\\-6\end{pmatrix}$ ⇐

$d(x_2) = \begin{pmatrix}1 & 2 & 1\end{pmatrix}\begin{pmatrix}10\\-10\\-6\end{pmatrix} = -16$ ⇒ Update Weight $\underline{W} = \begin{pmatrix}10\\-10\\-6\end{pmatrix} + 2 * \begin{pmatrix}1\\2\\1\end{pmatrix} = \begin{pmatrix}12\\-6\\-4\end{pmatrix}$ ⇐

$d(x_3) = \begin{pmatrix}1 & 1 & 2\end{pmatrix}\begin{pmatrix}12\\-6\\-4\end{pmatrix} = -2$ ⇒ Update Weight $\underline{W} = \begin{pmatrix}12\\-6\\-4\end{pmatrix} + 2 * \begin{pmatrix}1\\1\\2\end{pmatrix} = \begin{pmatrix}14\\-4\\0\end{pmatrix}$ ⇐

$d(x_4) = \begin{pmatrix}1 & 5 & 4\end{pmatrix}\begin{pmatrix}14\\-4\\0\end{pmatrix} = -6$ ⇒ Keep same Weight $\underline{W} = \begin{pmatrix}14\\-4\\0\end{pmatrix}$

$d(x_5) = \begin{pmatrix}1 & 6 & 5\end{pmatrix}\begin{pmatrix}14\\-4\\0\end{pmatrix} = -10$ ⇒ Keep same Weight $\underline{W} = \begin{pmatrix}14\\-4\\0\end{pmatrix}$

$d(x_6) = \begin{pmatrix}1 & 6 & 6\end{pmatrix}\begin{pmatrix}14\\-4\\0\end{pmatrix} = -112$ ⇒ Keep same Weight $\underline{W} = \begin{pmatrix}14\\-4\\0\end{pmatrix}$

| $x_1$ | $x_2$ | $x_3$ | $x_4$ | $x_5$ | $x_6$ |
|---|---|---|---|---|---|
| (1) | (2) | (1) | (5) | (6) | (6) |
| (1) | (1) | (2) | (4) | (5) | (6) |
| (1) | (1) | (1) | (4) | (5) | (6) |
| $C_1$ | $C_1$ | $C_1$ | $C_2$ | $C_2$ | $C_2$ |

Initial $\underline{W} = \begin{vmatrix} 4 \\ -2 \\ -2 \end{vmatrix}$ ***Step η = 2***

$C_1$ positive, $C_2$ negative, **Zero** is unaccepted

Check classes of

| $Z_1$ | $Z_1$ | $Z_1$ |
|---|---|---|
| (0) | (7) | (3) |
| (0) | (6) | (2) |

*Ugmented Vectors ( from Top )*

| | | | | | |
|---|---|---|---|---|---|
| (1) | (1) | (1) | (1) | (1) | (1) |
| (1) | (2) | (1) | (5) | (6) | (6) |
| (1) | (1) | (2) | (4) | (5) | (6) |
| (1) | (1) | (1) | (4) | (5) | (6) |
| $C_1$ | $C_1$ | $C_1$ | $C_2$ | $C_2$ | $C_2$ |

**Iteration 4**

**No change in $\underline{W}$**

***Stop Iterations***

$d(x_1) = \begin{vmatrix} 1 & 1 & 1 \end{vmatrix} \begin{vmatrix} 14 \\ -4 \\ 0 \end{vmatrix} = 10$ ⇒ Keep same Weight $\underline{W} = \begin{vmatrix} 14 \\ -4 \\ 0 \end{vmatrix}$

$d(x_2) = \begin{vmatrix} 1 & 2 & 1 \end{vmatrix} \begin{vmatrix} 14 \\ -4 \\ 0 \end{vmatrix} = 6$ ⇒ Keep same Weight $\underline{W} = \begin{vmatrix} 14 \\ -4 \\ 0 \end{vmatrix}$

$d(x_3) = \begin{vmatrix} 1 & 1 & 2 \end{vmatrix} \begin{vmatrix} 14 \\ -4 \\ 0 \end{vmatrix} = 10$ ⇒ Keep same Weight $\underline{W} = \begin{vmatrix} 14 \\ -4 \\ 0 \end{vmatrix}$

$d(x_4) = \begin{vmatrix} 1 & 5 & 4 \end{vmatrix} \begin{vmatrix} 14 \\ -4 \\ 0 \end{vmatrix} = -6$ ⇒ Keep same Weight $\underline{W} = \begin{vmatrix} 14 \\ -4 \\ 0 \end{vmatrix}$

$d(x_5) = \begin{vmatrix} 1 & 6 & 5 \end{vmatrix} \begin{vmatrix} 14 \\ -4 \\ 0 \end{vmatrix} = -10$ ⇒ Keep same Weight $\underline{W} = \begin{vmatrix} 14 \\ -4 \\ 0 \end{vmatrix}$

$d(x_6) = \begin{vmatrix} 1 & 6 & 6 \end{vmatrix} \begin{vmatrix} 14 \\ -4 \\ 0 \end{vmatrix} = -112$ ⇒ Keep same Weight $\underline{W} = \begin{vmatrix} 14 \\ -4 \\ 0 \end{vmatrix}$

$x_1$ $x_2$ $x_3$ $x_4$ $x_5$ $x_6$

$\begin{pmatrix}1\\1\end{pmatrix}$ $\begin{pmatrix}2\\1\end{pmatrix}$ $\begin{pmatrix}1\\2\end{pmatrix}$ $\begin{pmatrix}5\\4\end{pmatrix}$ $\begin{pmatrix}6\\5\end{pmatrix}$ $\begin{pmatrix}6\\6\end{pmatrix}$

$C_1$ $C_1$ $C_1$ $C_2$ $C_2$ $C_2$

Initial $\underline{W} = \begin{pmatrix}4\\-2\\-2\end{pmatrix}$ ***Step η = 2***

$C_1$ *positive* $C_2$ *negative* ***Zero*** *is unaccepted*

Check classes of $Z_1$ $Z_1$ $Z_1$

$\begin{pmatrix}0\\0\end{pmatrix}$ $\begin{pmatrix}7\\6\end{pmatrix}$ $\begin{pmatrix}3\\2\end{pmatrix}$

$\underline{W} = \begin{pmatrix}14\\-4\\0\end{pmatrix}$

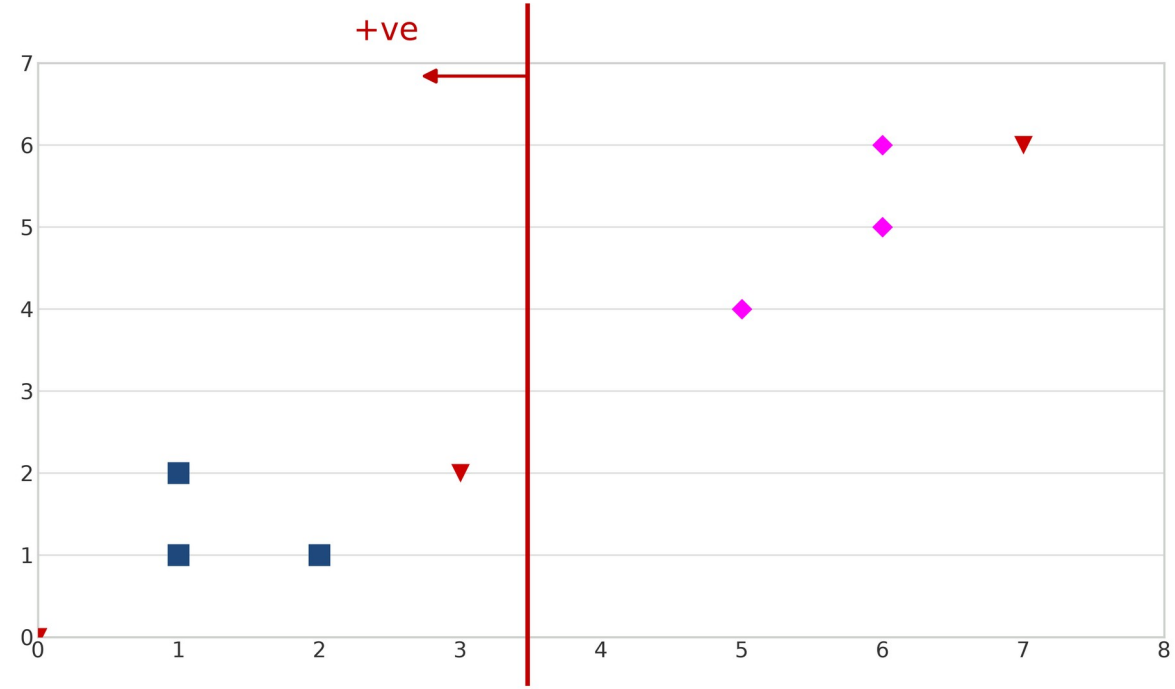
***Ugmented Vectors ( from Top )***

$x_1$ $x_2$ $x_3$ $x_4$ $x_5$ $x_6$

$\begin{pmatrix}1\\1\\1\end{pmatrix}$ $\begin{pmatrix}1\\2\\1\end{pmatrix}$ $\begin{pmatrix}1\\1\\2\end{pmatrix}$ $\begin{pmatrix}1\\5\\4\end{pmatrix}$ $\begin{pmatrix}1\\6\\5\end{pmatrix}$ $\begin{pmatrix}1\\6\\6\end{pmatrix}$

$C_1$ $C_1$ $C_1$ $C_2$ $C_2$ $C_2$

## Iteration 4




No change in ***W***

***Stop*** *Iterations*

| $x_1$ | $x_2$ | $x_3$ | $x_4$ | $x_5$ | $x_6$ |
|---|---|---|---|---|---|
| 1 | 2 | 1 | 5 | 6 | 6 |
| 1 | 1 | 2 | 4 | 5 | 6 |
| $C_1$ | $C_1$ | $C_1$ | $C_2$ | $C_2$ | $C_2$ |

Initial $\underline{\mathbf{W}}$ = $\begin{vmatrix} 4 \\ -2 \\ -2 \end{vmatrix}$

***Step η = 2***

Check classes of

| $Z_1$ | $Z_1$ | $Z_1$ |
|---|---|---|
| 0 | 7 | 3 |
| 0 | 6 | 2 |

$C_1$ positive $C_2$negative Zero is unaccepted

*Ugmented Vectors ( from Top )*

| $x_1$ | $x_2$ | $x_3$ | $x_4$ | $x_5$ | $x_6$ |
|---|---|---|---|---|---|
| 1 | 1 | 1 | 1 | 1 | 1 |
| 1 | 2 | 1 | 5 | 6 | 6 |
| 1 | 1 | 2 | 4 | 5 | 6 |
| $C_1$ | $C_1$ | $C_1$ | $C_2$ | $C_2$ | $C_2$ |

## Check Classes

$d(Z_1)$ = $\begin{vmatrix} 1 & 0 & 0 \end{vmatrix} \begin{vmatrix} 14 \\ -4 \\ 0 \end{vmatrix}$ = 14 ⇒ **Positive > 0** **Class I**

$d(Z_2)$ = $\begin{vmatrix} 1 & 7 & 6 \end{vmatrix} \begin{vmatrix} 14 \\ -4 \\ 0 \end{vmatrix}$ = -14 ⇒ **Negative < 0** **Class II**

$d(Z_2)$ = $\begin{vmatrix} 1 & 3 & 2 \end{vmatrix} \begin{vmatrix} 14 \\ -4 \\ 0 \end{vmatrix}$ = -2 ⇒ **Negative < 0** **Class II**

| $x_1$ | $x_2$ | $x_3$ | $x_4$ | $x_5$ | $x_6$ |
|---|---|---|---|---|---|
| (1) | (2) | (1) | (5) | (6) | (6) |
| (1) | (1) | (2) | (4) | (5) | (6) |
| C1 | C1 | C1 | C2 | C2 | C2 |

$\begin{vmatrix} 4 \\ -2 \\ -2 \end{vmatrix}$ *Initial* $W$ = **REPEAT using Step η = 1**

$C_1$ positive $C_2$ negative **Zero** is unaccepted

Check classes of $Z_1 \begin{vmatrix} 0 \\ 0 \end{vmatrix}$ $Z_1 \begin{vmatrix} 7 \\ 6 \end{vmatrix}$ $Z_1 \begin{vmatrix} 3 \\ 2 \end{vmatrix}$

*Ugmented Vectors (from Top)*

| $x_1$ | $x_2$ | $x_3$ | $x_4$ | $x_5$ | $x_6$ |
|---|---|---|---|---|---|
| (1) | (1) | (1) | (1) | (1) | (1) |
| (1) | (2) | (1) | (5) | (6) | (6) |
| (1) | (1) | (2) | (4) | (5) | (6) |
| $C_1$ | $C_1$ | $C_1$ | $C_2$ | $C_2$ | $C_2$ |

**Iteration 1**

$d(x_1) = \begin{vmatrix} 1 & 1 & 1 \end{vmatrix} \cdot \begin{vmatrix} 4 \\ -2 \\ -2 \end{vmatrix} = 0$ ⇒ Update Weight $\underline{W} = \begin{vmatrix} 4 \\ -2 \\ -2 \end{vmatrix} + 1 * \begin{vmatrix} 1 \\ 1 \\ 1 \end{vmatrix} = \begin{vmatrix} 5 \\ -1 \\ -1 \end{vmatrix}$ ⇐

$d(x_2) = \begin{vmatrix} 1 & 2 & 1 \end{vmatrix} \cdot \begin{vmatrix} 5 \\ -1 \\ -1 \end{vmatrix} = 2$ ⇒ Keep same Weight $\underline{W} = \begin{vmatrix} 5 \\ -1 \\ -1 \end{vmatrix}$

$d(x_3) = \begin{vmatrix} 1 & 1 & 2 \end{vmatrix} \cdot \begin{vmatrix} 5 \\ -1 \\ -1 \end{vmatrix} = 2$ ⇒ Keep same Weight $\underline{W} = \begin{vmatrix} 5 \\ -1 \\ -1 \end{vmatrix}$

$d(x_4) = \begin{vmatrix} 1 & 5 & 4 \end{vmatrix} \cdot \begin{vmatrix} 5 \\ -1 \\ -1 \end{vmatrix} = -4$ ⇒ Keep same Weight $\underline{W} = \begin{vmatrix} 5 \\ -1 \\ -1 \end{vmatrix}$

$d(x_5) = \begin{vmatrix} 1 & 6 & 5 \end{vmatrix} \cdot \begin{vmatrix} 5 \\ -1 \\ -1 \end{vmatrix} = -6$ ⇒ Keep same Weight $\underline{W} = \begin{vmatrix} 5 \\ -1 \\ -1 \end{vmatrix}$

$d(x_6) = \begin{vmatrix} 1 & 6 & 6 \end{vmatrix} \cdot \begin{vmatrix} 5 \\ -1 \\ -1 \end{vmatrix} = -7$ ⇒ Keep same Weight $\underline{W} = \begin{vmatrix} 5 \\ -1 \\ -1 \end{vmatrix}$

| $x_1$ | $x_2$ | $x_3$ | $x_4$ | $x_5$ | $x_6$ |
|---|---|---|---|---|---|
| 1 | 2 | 1 | 5 | 6 | 6 |
| 1 | 1 | 2 | 4 | 5 | 6 |
| $C_1$ | $C_1$ | $C_1$ | $C_2$ | $C_2$ | $C_2$ |

$$\begin{bmatrix}4\\-2\\-2\end{bmatrix}$$ *Initial W* **REPEAT using Step η = 1**

$C_1$ positive $C_2$ negative **Zero** is unaccepted

Check classes of

| $Z_1$ | $Z_2$ | $Z_3$ |
|---|---|---|
| 0 | 7 | 3 |
| 0 | 6 | 2 |

*Ugmented Vectors (from Top)*

| $x_1$ | $x_2$ | $x_3$ | $x_4$ | $x_5$ | $x_6$ |
|---|---|---|---|---|---|
| 1 | 1 | 1 | 1 | 1 | 1 |
| 1 | 2 | 1 | 5 | 6 | 6 |
| 1 | 1 | 2 | 4 | 5 | 6 |
| $C_1$ | $C_1$ | $C_1$ | $C_2$ | $C_2$ | $C_2$ |

## Iteration 2

No change in W
*Stop* Iterations

$d(x_1) = \begin{bmatrix}1&1&1\end{bmatrix}\begin{bmatrix}5\\-1\\-1\end{bmatrix} = 3$ ⇒ Keep same Weight $\underline{W} = \begin{bmatrix}5\\-1\\-1\end{bmatrix}$

$d(x_2) = \begin{bmatrix}1&2&1\end{bmatrix}\begin{bmatrix}5\\-1\\-1\end{bmatrix} = 2$ ⇒ Keep same Weight $\underline{W} = \begin{bmatrix}5\\-1\\-1\end{bmatrix}$

$d(x_3) = \begin{bmatrix}1&1&2\end{bmatrix}\begin{bmatrix}5\\-1\\-1\end{bmatrix} = 2$ ⇒ Keep same Weight $\underline{W} = \begin{bmatrix}5\\-1\\-1\end{bmatrix}$

$d(x_4) = \begin{bmatrix}1&5&4\end{bmatrix}\begin{bmatrix}5\\-1\\-1\end{bmatrix} = -4$ ⇒ Keep same Weight $\underline{W} = \begin{bmatrix}5\\-1\\-1\end{bmatrix}$

$d(x_5) = \begin{bmatrix}1&6&5\end{bmatrix}\begin{bmatrix}5\\-1\\-1\end{bmatrix} = -6$ ⇒ Keep same Weight $\underline{W} = \begin{bmatrix}5\\-1\\-1\end{bmatrix}$

$d(x_6) = \begin{bmatrix}1&6&6\end{bmatrix}\begin{bmatrix}5\\-1\\-1\end{bmatrix} = -7$ ⇒ Keep same Weight $\underline{W} = \begin{bmatrix}5\\-1\\-1\end{bmatrix}$

| $x_1$ | $x_2$ | $x_3$ | $x_4$ | $x_5$ | $x_6$ |
|---|---|---|---|---|---|
| 1 | 2 | 1 | 5 | 6 | 6 |
| 1 | 1 | 2 | 4 | 5 | 6 |
| $C_1$ | $C_1$ | $C_1$ | $C_2$ | $C_2$ | $C_2$ |

Initial $\boldsymbol{W} = \begin{pmatrix} 4 \\ -2 \\ -2 \end{pmatrix}$ *REPEAT* ***using Step η = 1***

$C_1$ positive $C_2$ negative Zero is unaccepted

Check classes of $Z_1 \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ $Z_2 \begin{pmatrix} 7 \\ 6 \end{pmatrix}$ $Z_3 \begin{pmatrix} 3 \\ 2 \end{pmatrix}$

$\underline{W} = \begin{pmatrix} 5 \\ -1 \\ -1 \end{pmatrix}$

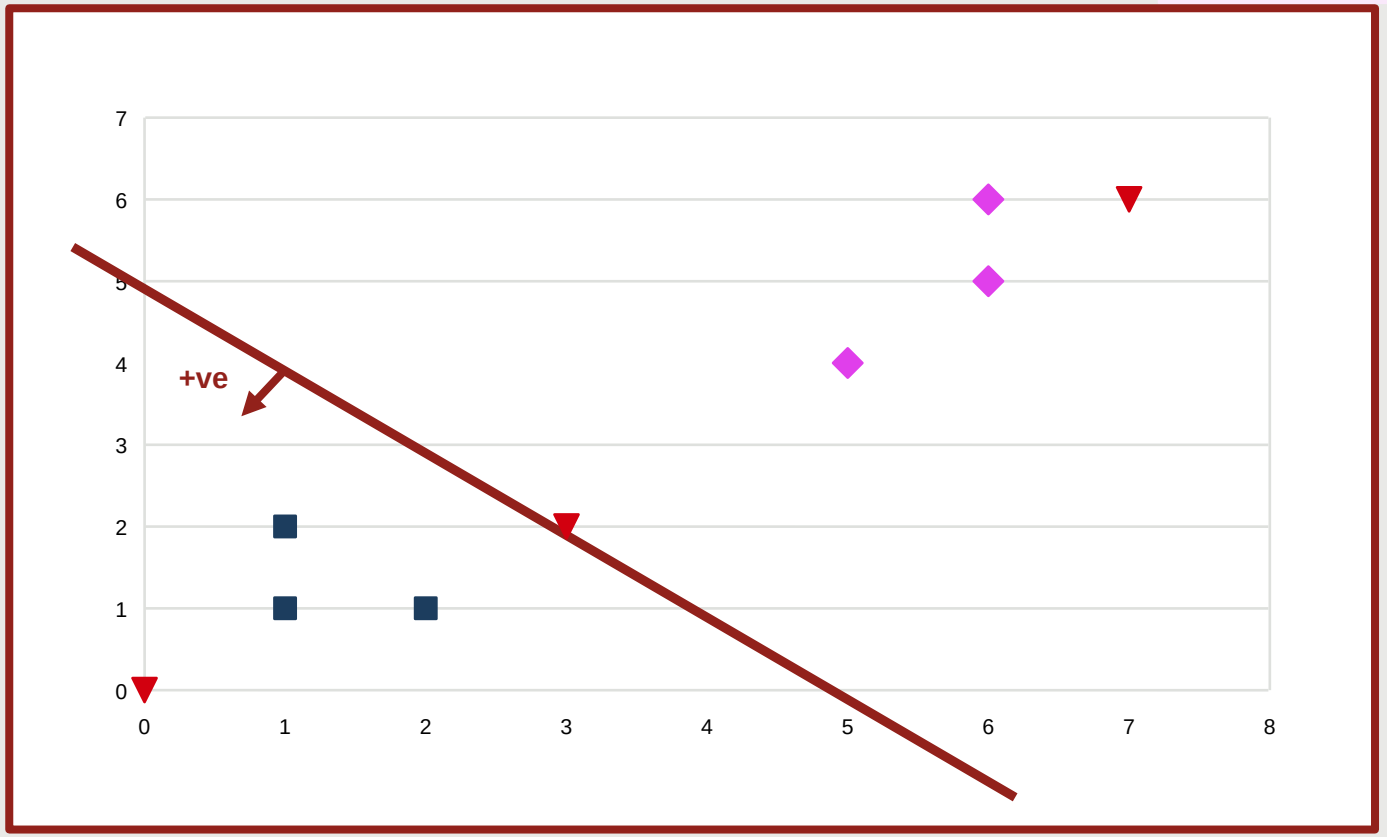
**Augmented Vectors (from Top)**

| $x_1$ | $x_2$ | $x_3$ | $x_4$ | $x_5$ | $x_6$ |
|---|---|---|---|---|---|
| 1 | 1 | 1 | 1 | 1 | 1 |
| 1 | 2 | 1 | 5 | 6 | 6 |
| 1 | 1 | 2 | 4 | 5 | 6 |
| $C_1$ | $C_1$ | $C_1$ | $C_2$ | $C_2$ | $C_2$ |

**Iteration 2**

No change in $\underline{W}$

**Stop** Iterations



$x_1 \; x_2 \; x_3 \; x_4 \; x_5 \; x_6$

$\begin{pmatrix}1\\1\end{pmatrix} \begin{pmatrix}2\\1\end{pmatrix} \begin{pmatrix}1\\2\end{pmatrix} \begin{pmatrix}5\\4\end{pmatrix} \begin{pmatrix}6\\5\end{pmatrix} \begin{pmatrix}6\\6\end{pmatrix}$

$C_1 \; C_1 \; C_1 \; C_2 \; C_2 \; C_2$

Initial $\boldsymbol{W} = \begin{pmatrix}4\\-2\\-2\end{pmatrix}$ $Step\,\eta = 1$

Check classes of $Z_1 \; Z_2 \; Z_3$ $\begin{pmatrix}0\\0\end{pmatrix} \begin{pmatrix}7\\6\end{pmatrix} \begin{pmatrix}3\\2\end{pmatrix}$

$C_1$ *positive* $C_2$ *negative* ***Zero*** is unaccepted

***Ugmented Vectors ( from Top )***

$x_1 \; x_2 \; x_3 \; x_4 \; x_5 \; x_6$

$\begin{pmatrix}1\\1\\1\end{pmatrix} \begin{pmatrix}1\\2\\1\end{pmatrix} \begin{pmatrix}1\\1\\2\end{pmatrix} \begin{pmatrix}1\\5\\4\end{pmatrix} \begin{pmatrix}1\\6\\5\end{pmatrix} \begin{pmatrix}1\\6\\6\end{pmatrix}$

$C_1 \; C_1 \; C_1 \; C_2 \; C_2 \; C_2$

**Check Classes**

$d(Z_1) = (1\ 0\ 0)\begin{pmatrix}5\\-1\\-1\end{pmatrix} = 5$ ⇒ Positive > 0 **Class I**

$d(Z_2) = (1\ 7\ 6)\begin{pmatrix}5\\-1\\-1\end{pmatrix} = -8$ ⇒ Negative < 0 **Class II**

$d(Z_2) = (1\ 3\ 2)\begin{pmatrix}5\\-1\\-1\end{pmatrix} = 0$ ⇒ Not > 0 , Not < 0 **Undefined Class**